An AI Agent Framework for Autism Behavioral Trait Detection in Children

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A Capstone Project Report submitted in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering



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Declaration

We, Sharafat Ahmed, Rajon Das, Jannatul Ferdous, and Lima Prodhan hereby, declare that this capstone project report entitled "An AI Agent Framework for Autism Behavioral Trait Detection in Children" is our original work and has been carried out under the supervision of **Md AI Imran**, Senior Lecturer, Department of Computer Science and Engineering, East West University. We further declare that this work has not been submitted anywhere else for the award of any degree, diploma, or other academic qualification.

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Letter of Acceptance

The capstone project report entitled "An AI Agent Framework for Autism Behavioral Trait Detection in Children" is submitted by Sharafat Ahmed, Rajon Das, Jannatul Ferdous, and Lima Prodhan to the Department of Computer Science and Engineering, East West University, Dhaka, Bangladesh and is accepted for the partial fulfillment of the requirement for the degree of Bachelor of Science in Computer Science and Engineering.

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Abstract

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition characterised by persistent deficits in social communication and restricted, repetitive patterns of behaviour. Despite the availability of validated screening instruments, early identification remains a global challenge due to limited access to specialists, high diagnostic costs, and the absence of scalable, caregiver-friendly tools. This project presents an AI Agent Framework for Autism Trait Detection in Children, designed as an end-to-end conversational screening and support system.

The proposed framework integrates two independently trained machine learning models within a multi-agent orchestration architecture. A gradient-boosted decision tree model (XGBoost) is employed for structured questionnaire-based screening using Q-CHAT-10 features, achieving 98.6% test accuracy and near-perfect discriminative performance. A fine-tuned transformer-based model (BERT) is used for free-text behavioural analysis, achieving 88% accuracy and strong generalisation under data-constrained conditions through semantically conservative augmentation techniques.

Beyond classification, the system incorporates a DSM-5-aligned behavioural trait taxonomy, a concern-level scoring mechanism, and companion support features including personalised social story generation and visual schedule creation. A strict architectural boundary ensures that all diagnostic outputs are generated exclusively by validated machine learning models, while the language model is restricted to explanation, routing, and caregiver interaction.

The dual-modality design significantly enhances accessibility by accommodating both structured and narrative caregiver inputs. The proposed system demonstrates the potential of combining machine learning and agent-based AI to deliver scalable, low-cost, and accessible early ASD screening and support tools, particularly in resource-constrained settings.

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Chapter 1: Introduction

1.1 Background, Overview, and Motivation

Autism Spectrum Disorder is a heterogeneous neurodevelopmental condition defined in the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5) by two core symptom domains: persistent deficits in social communication and social interaction across multiple contexts, and restricted, repetitive patterns of behaviour, interests, or activities [1]. Its aetiology is multifactorial, involving complex interactions between genetic predisposition and environmental influences, and its clinical presentation is highly variable across the lifespan. The term "spectrum" reflects this heterogeneity: individuals with ASD range from those who are non-verbal with severe intellectual disability to those with high intellectual ability and subtle social difficulties.

The epidemiology of ASD has shifted markedly over recent decades. The Autism and Developmental Disabilities Monitoring (ADDM) Network, maintained by the United States Centres for Disease Control and Prevention (CDC), reported in 2025 that ASD prevalence among eight-year-old children in 2022 was 32.2 per 1,000, or approximately one in 31, a significant increase from earlier surveillance cycles and the highest figure reported in the programme's history [7]. The global burden is similarly substantial; the World Health Organisation revised its global prevalence estimate in 2022 to approximately one in 100 children. This rising prevalence, combined with evidence that intensive early behavioural intervention in the first three years of life yields substantially better language, cognitive, and adaptive outcomes than later intervention, has generated a compelling public health argument for scalable early screening infrastructure [6].

The principal challenge is not awareness of ASD but access to timely screening and diagnosis. In high-income countries, the median age of earliest known ASD diagnosis remains approximately 47 months, nearly four years of age and despite the fact that reliable diagnostic assessment is clinically feasible in children as young as 18 to 24 months [7, 62]. This diagnostic lag is driven by multiple structural factors: a severe shortage of paediatric developmental specialists relative to population demand, long waiting lists for formal psychological assessment, high cost of comprehensive evaluation, geographic concentration of specialist services in urban centres, and cultural stigma that discourages early help-seeking in many communities. In low-and-middle-income countries, including many in South and South-East Asia, these barriers are more acute, and large proportions of affected children receive no professional assessment during the critical early intervention window.

A parallel development in artificial intelligence, specifically, the maturation of pre-trained transformer language models and gradient-boosted tree classifiers has created a practical opportunity to address this infrastructure gap. Validated behavioural screening instruments

such as the Q-CHAT-10 generate structured binary feature vectors that are highly amenable to tabular machine learning classification [3]. Caregiver narratives describing child behaviour constitute free-text input that fine-tuned BERT models can classify with meaningful accuracy even from small training corpora [8, 9, 10]. And conversational AI agents, orchestrated through frameworks such as LangGraph, can present these ML capabilities to non-specialist caregivers in an accessible, guided, multi-turn interface that requires no clinical training to use [2].

This project is motivated by the convergence of these three developments. The clinical need for scalable early screening is clear and urgent. The ML tools to operationalise validated screening instruments are available and mature. The conversational agent infrastructure to package those tools for caregiver use has recently become technically feasible. What has been missing is a system that integrates all three in a principled, architecturally safe manner, one that preserves the validated ML diagnostic boundary while providing the warm, contextual caregiver support that makes a screening tool genuinely useful rather than merely technically correct.

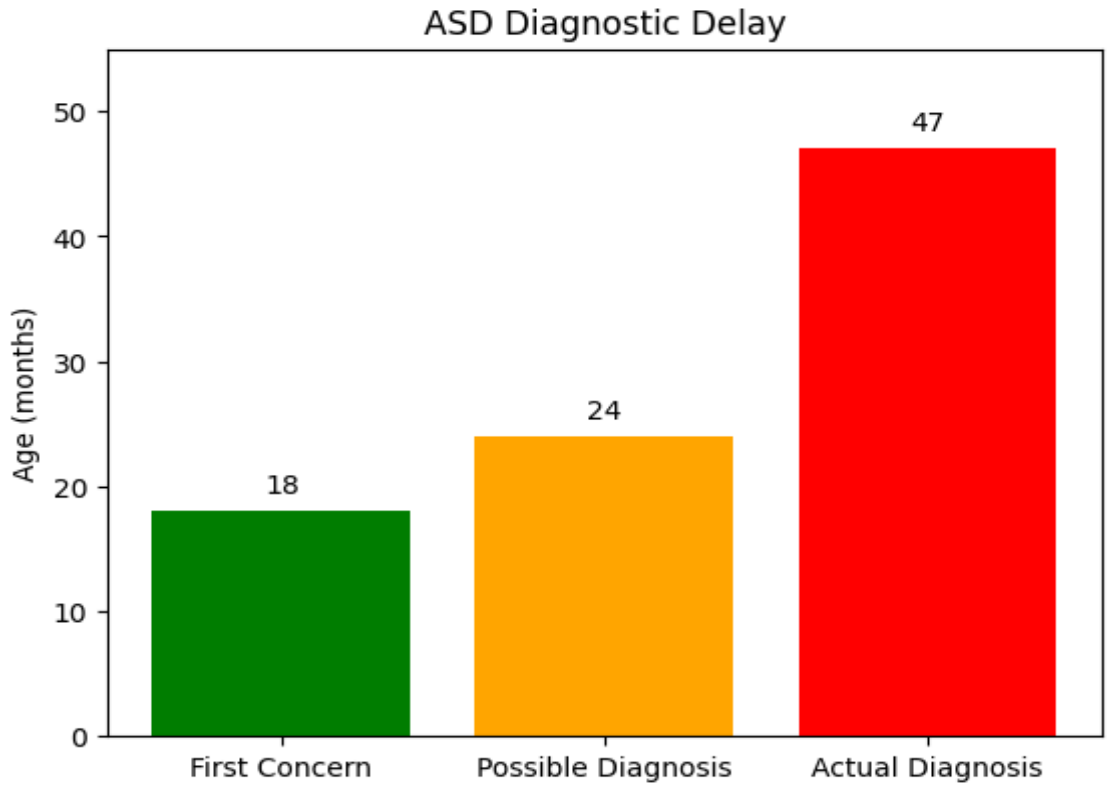


Figure 1: Bar chart showing the diagnostic age gap in months

1.2 Problem Statement and Analysis

Current approaches to ASD screening and early detection fall into three broad categories, each with significant limitations that this project addresses.

The first category is formal clinical assessment. Comprehensive evaluation by a developmental paediatrician, child psychiatrist, or clinical psychologist using standardised instruments such as the Autism Diagnostic Observation Schedule, Second Edition (ADOS-2) or the Autism Diagnostic Interview-Revised (ADI-R) represents the diagnostic gold standard. However, this approach is expensive (assessments typically cost hundreds to thousands of dollars), scarce (wait times of six to eighteen months are common in many health systems), geographically concentrated, and entirely inaccessible to families without insurance or specialist proximity. It cannot function as population-level screening.

The second category consists of validated paper-based and digital screening instruments, including the Q-CHAT-10 for toddlers and the M-CHAT-R/F for slightly older children. These instruments are brief, validated, and can be administered by primary care providers or parents. However, they remain confined largely to clinical settings, they are purely binary instruments that return a score rather than an explanation, they do not accommodate the large proportion of caregivers who prefer narrative description to structured question-answering, and they provide no post-result guidance or companion support.

The third category consists of emerging research-stage ML classifiers for ASD detection, which have proliferated in recent years. The literature demonstrates that XGBoost, Random Forest, and BERT-based models can achieve high accuracy on tabular and text ASD datasets [5, 8, 9, 10]. However, these systems remain isolated research artefacts: they are not deployed in conversational interfaces, they do not support both input modalities simultaneously, they do not maintain cross-turn conversational context, they have no companion capabilities, and they frequently suffer from methodological problems such as data leakage (specifically, including the total Q-CHAT score as a predictive feature) and inadequate cross-validation [5].

A specific and underappreciated gap in all existing approaches is the modality assumption: every validated instrument and every published ML classifier assumes a single, fixed input format. Either the caregiver answers structured questions, or the caregiver provides free text but not both. In practice, many caregivers are better able to describe their child's behaviour in natural language than to map those observations onto a binary questionnaire. This project is the first to address both modalities in a single deployed system.

A further critical gap is the total absence of companion functionality in existing tools. Screening and diagnosis are not terminal events; they initiate a long process of understanding, therapy, and daily adaptation. Parents who receive a screening result need to understand what it means, what specific behaviours were identified, what they can do at home, and how to prepare their child for challenging situations. No existing screening tool provides any of this.

This project addresses the gap through social story generation, visual schedule generation, and contextual caregiver guidance.

Finally, all existing conversational AI systems that engage with health topics face the hallucination risk documented by Thirunavukarasu et al. [2] and empirically characterised by Umapathi et al. [11]: language models generate plausible-sounding but factually incorrect medical statements at rates sufficient to cause harm in clinical decision-making contexts. No existing ASD screening system has implemented an architectural anti-hallucination boundary. This project does so by construction, through strict separation of ML-generated diagnostic outputs from LLM-generated conversational outputs.

1.3 Project Objectives

The specific objectives of this project are as follows:

Objective 1. To develop and validate an XGBoost classifier that accepts the eleven binary features of the Q-CHAT-10 instrument as input and produces a calibrated ASD/Non-ASD prediction with probability scores, outperforming Random Forest, KNN, and Logistic Regression baselines on the same dataset.

Objective 2. To develop and validate a fine-tuned BERT classifier that accepts free-text caregiver behavioural descriptions and performs binary ASD classification, with semantically conservative data augmentation applied to triple the effective training corpus while preserving clinical vocabulary.

Objective 3. To design and implement a LangGraph multi-agent orchestration architecture that routes conversational input to the appropriate ML classifier, enforces an architectural anti-hallucination boundary, and maintains persistent multi-turn conversational state via MemorySaver checkpointing.

Objective 4. To implement a 14-trait behavioural taxonomy module grounded in DSM-5 social-communication and restricted behaviour domains, and to compute a three-tier concern severity score from trait-level analysis.

Objective 5. To implement companion capability modules as personalised social story generation with FAISS semantic retrieval, visual schedule generation, and child profile management that extend the system from a one-shot screening instrument to an ongoing caregiver companion.

Objective 6. To construct a Streamlit conversational interface that integrates all components and presents results in a format accessible to non-specialist caregivers, with real-time child profile display, session history tracking, and concern-level visual badges.

1.4 Project Contributions

The primary research and engineering contributions of this project are as follows.

Contribution 1 : Dual-modality ASD screening agent.

This project delivers the first conversational agent that simultaneously supports structured questionnaire-based and free-text behavioural description-based ASD screening in a single session, with the user choosing the modality that best suits their situation and cognitive style.

Contribution 2 : Architectural anti-hallucination boundary.

The system enforces a formal, architectural separation between ML diagnostic outputs and LLM conversational outputs. No classification statement is generated by the language model; all diagnosis-adjacent outputs are produced exclusively by the trained XGBoost and BERT classifiers. This is a principled engineering response to the hallucination risk in medical AI [2, 11].

Contribution 3 : 14-trait taxonomy with DSM-5-aligned severity scoring.

A structured 14-trait behavioural taxonomy, grounded in the two core DSM-5 diagnostic domains, is implemented and applied to convert free-text screening results into interpretable, trait-level explanatory output with a three-tier concern severity score (High, Moderate, Monitor).

Contribution 4 : Semantically conservative BERT fine-tuning under data scarcity.

A two-pass augmentation strategy which is protected-vocabulary WordNet synonym replacement and contextual BERT MLM augmentation is implemented and validated for BERT fine-tuning on a 332-sample clinical text corpus, demonstrating that clinical semantic fidelity can be maintained while tripling the effective training set size.

Contribution 5 : Companion capabilities with FAISS retrieval.

The framework integrates social story generation, visual schedule generation, FAISS-based semantic retrieval of pre-built materials, sensory trigger detection, and session history tracking, transforming a diagnostic screening tool into an ongoing caregiver companion.

1.5 Project Outline

The remainder of this report is structured as follows:

- **Chapter 2** surveys the state-of-the-art across ML-based ASD screening, NLP and transformer models in healthcare, multi-agent AI orchestration, and evidence-based companion interventions, closing with a gap-analysis summary table.

- **Chapter 3** presents the methodology: datasets, preprocessing, augmentation, model architectures, the multi-agent framework design, and the implementation environment.

- **Chapter 4** reports and analyses the experimental results for both ML models and the integrated system.

- **Chapter 5** presents screenshots from the deployed Streamlit web application to illustrate the system's conversational flow and user interface design.

- **Chapter 6** concludes with a summary of contributions, limitations, future directions, and a business model analysis.

Chapter 2: Related Work

2.1 Survey of the State-of-the-Art

2.1.1 Machine Learning for ASD Detection

Machine learning applied to structured ASD screening data has been an active and productive research area for over a decade. The foundational survey by Thabtah [5] systematically reviewed the application of decision trees, k-nearest neighbours, naive Bayes, and neural networks across multiple age-group ASD datasets collected via the AutismTests mobile application, establishing the Q-CHAT-10 toddler dataset as a benchmark for ML ASD classification research. Thabtah's analysis also identified a persistent methodological problem across the literature: many published classifiers inadvertently included the total Q-CHAT score as a feature, which is the arithmetic sum of the ten questionnaire items and therefore a trivially perfect predictor of the target. This data leakage inflated reported accuracies and rendered results incomparable across studies. The present project explicitly removes this feature before any model fitting.

Tartarisco et al. [12] applied five ML algorithms (Random Forest, Naive Bayes, SVM, Logistic Regression, KNN) to the full Q-CHAT instrument which contains 25 items rather than the 10-item screener and reported an overall accuracy of 90%, with SVM achieving 95%. Their work demonstrated that ML can capture discriminative structure in Q-CHAT item responses beyond what the simple sum score reveals. However, the full Q-CHAT is more burdensome to administer than the Q-CHAT-10, reducing practical applicability in population screening contexts.

More recently, a multicultural study published in 2024 trained Decision Tree, Random Forest, and XGBoost models on Q-CHAT-10 datasets from three countries: Poland, New Zealand, and Saudi Arabia and tested cross-cultural transfer [13]. XGBoost consistently outperformed both alternatives, achieving AUROCs of 0.94 across the multicultural transfer condition. This finding corroborates the XGBoost selection in the present project and supports the argument that Q-CHAT-10-based ML classifiers generalise across cultural contexts. Importantly, the study also found that a compact four-feature subset retained most of the discriminative power, suggesting that future work on the Q-CHAT-10 should investigate minimal-feature variants for deployment in very low-resource settings.

Chen and Guestrin [14] introduced XGBoost as a scalable, regularised gradient-boosted tree framework that consistently outperforms other ensemble methods on tabular classification benchmarks. Its key advantages for the ASD screening context include the scale_pos_weight parameter for handling class imbalance without resampling, built-in L1 and L2 regularisation to prevent overfitting on small-to-medium datasets, and complete scale-invariance that

eliminates the need for feature standardisation when all inputs are binary. A comprehensive recent review covering ASD detection from questionnaire, imaging, facial, EEG, and eye-tracking data confirmed that gradient-boosted methods including XGBoost dominate accuracy rankings on questionnaire-based ASD datasets [15].

2.1.2 Natural Language Processing and Deep Learning for ASD Screening

The application of deep learning and transformer models to ASD detection from text represents a more recent and rapidly evolving research direction. The release of BERT by Devlin et al. [16] established bidirectional contextual pre-training as the state of the art for a wide range of NLP classification tasks. In the clinical domain, a systematic review by Thirunavukarasu et al. [2] confirmed that large language models perform at or near specialist level across numerous medical benchmarks while documenting the hallucination risk that makes safety-critical architectural design essential in clinical deployment.

Pham et al. [8] demonstrated that BERT models applied to social media text from individuals self-identified as having ASD could achieve approximately 88% classification accuracy, directly comparable to the performance of the present project's BERT modality on a clinically curated dataset. Their study used BERTweet, a variant pre-trained on Twitter data, as the top-performing model. The convergence of performance across different text sources (social media vs. caregiver narrative) and different BERT variants suggests that the underlying distributional signatures of ASD-associated language are robust across genres.

Mukherjee et al. [9] applied BERT and ChatGPT models to parent dialogue about autistic children, achieving 83% accuracy with BERT and demonstrating that conversational text from caregivers carries sufficient discriminative signal for ASD detection. Their work is directly relevant to the present project's use case caregiver narrative as input and supports the design decision to train on free-text behavioural descriptions rather than structured questionnaire responses.

A study combining BERT-based narrative analysis with traditional NLP linguistic feature extraction achieved strong ASD classification accuracy on speech transcripts from children, showing that semantic embeddings from BERT capture qualitatively different and complementary information to manually engineered linguistic features [17]. This motivates the present project's use of BERT embeddings over a simpler bag-of-words or TF-IDF approach: the bidirectional attention mechanism captures semantic context that is critical for interpreting clinical behavioural descriptions.

Wawer et al. [18] conducted cross-dataset BERT fine-tuning experiments for ASD detection from transcribed speech, finding that data augmentation helps alleviate the small-sample problem but that cross-dataset transfer performance is task-specific. This directly motivates the augmentation strategy in the present project. Wei and Zou [4] demonstrated in a seminal empirical study that synonym replacement is the single most effective simple augmentation

technique for text classification in low-resource settings that outperforming random insertion, random swap, and random deletion which providing the primary justification for incorporating synonym augmentation as the first pass in the present project's two-pass strategy.

2.1.3 Multi-Agent AI Systems and Conversational Health Agents

The orchestration of multiple specialised AI agents within a shared state graph represents the current frontier of applied LLM engineering. LangGraph, introduced by LangChain in 2023 as a graph-based extension of the LangChain framework, provides the foundational infrastructure for building stateful multi-agent workflows with cyclical control flow, conditional routing, and persistent memory via MemorySaver checkpointing [19]. Unlike earlier pipeline frameworks that assumed acyclic data flow, LangGraph supports the multi-turn conversational pattern essential for clinical interview-style interactions, in which the agent must both remember previous responses and dynamically re-route based on user intent.

Duan et al. [20] explored the integration of LangGraph with CrewAI for complex multi-agent task management, demonstrating that the combination of LangGraph's graph-based control flow with specialised role-based agent assignment enables reliable execution of workflows that involve iterative reasoning and state-dependent branching. Their framework shares architectural features with the present project's supervisor-specialist agent design, where the supervisor performs intent classification and the specialist agents perform domain-specific tasks.

The application of conversational AI to healthcare and mental health support has grown substantially. The specific risk of LLM hallucination in clinical contexts has been empirically documented by multiple research groups [2, 11, 21]. Umapathi et al. [11] showed that even medically purpose-built LLMs remain vulnerable to domain-specific hallucinations arising from reasoning failures rather than mere knowledge gaps, highlighting the necessity of architectural guardrails rather than reliance on model-level safety mechanisms. The anti-hallucination boundary in the present project does routing all diagnostic output through ML classifiers and prohibiting the LLM from generating diagnostic statements is a direct engineering response to this literature.

Retrieval-Augmented Generation (RAG) has emerged as the dominant paradigm for grounding LLM outputs in verified domain knowledge. Gao et al. [22] provide a comprehensive survey of RAG architectures, identifying dense vector retrieval using sentence-transformer embeddings stored in FAISS indices as the most widely adopted and performant retrieval approach for knowledge-intensive generation tasks. Reimers and Gurevych [23] introduced Sentence-BERT (SBERT), demonstrating that the all-MiniLM-L6-v2 model provides an optimal trade-off between semantic representation

quality and computational efficiency for similarity search, which is the model adopted in the present project's retrieval engine.

Johnson et al. [24] describe the FAISS library, which implements efficient approximate nearest-neighbour search over dense embedding vectors, enabling sub-millisecond retrieval over libraries of thousands to millions of documents. The application of FAISS to a library of ASD social stories and visual schedules in the present project represents a novel use of the technology in the ASD assistive technology domain, enabling personalised material retrieval rather than pure LLM generation.

2.2.4 Evidence-Based Companion Interventions for ASD

Social stories, introduced by Gray and Garand [25] in 1993, are first-person narratives that describe social situations in structured, simple sentences, designed to improve comprehension and reduce anxiety before novel or challenging events. A comprehensive scoping review covering 56 empirical articles published between 2007 and 2018 confirmed that social story interventions are effective across a range of target behaviours in ASD populations, including reducing aggression, improving social interaction, increasing compliance with routines, and enhancing emotional regulation [26]. The review found that most studies were effective, with particularly strong evidence for implementation in home and school settings precisely the contexts in which the present project's AI-generated social stories would be used.

Visual activity schedules are evidence-based tools that present daily routines as ordered sequences of images or text steps, reducing transition-related distress and improving independent task completion in children with ASD [27]. Schneider and Goldstein [28] demonstrated that combining visual schedules with social story interventions produces larger effect sizes than either intervention alone ($d = 1.7$ combined vs. $d = 1.33$ for social story alone), which motivates the present project's integration of both capabilities within the same companion layer. The automated generation of personalised social stories and visual schedules from LLM prompting, constrained by child profile information and semantic retrieval of similar pre-built examples, represents a practical translation of these clinical tools into the AI domain.

The child profile system implemented in this project capturing the child's name, age, known sensory triggers, and current therapeutic goals enables the personalisation of all generated content, an approach grounded in the individualised intervention principle that underlies evidence-based ASD practice. Sensory trigger detection, which scans generated content for words associated with known triggers, further ensures that AI-generated materials are safe and appropriate for the specific child.

2.2 Summary of Research Gap

The literature review reveals several important gaps that motivate the present work. First, most existing ASD screening systems are limited to a single input modality, typically structured questionnaires, which restricts accessibility for diverse user populations. Second, while NLP techniques have shown promise in healthcare applications, their use in ASD screening remains constrained by limited data availability and the absence of robust multimodal integration.

Third, current AI-based screening tools often lack mechanisms to ensure reliability, particularly when incorporating large language models. The risk of hallucination in clinical contexts necessitates architectural safeguards that are not commonly implemented in existing systems. Finally, most tools focus solely on classification, without providing ongoing support or intervention resources for caregivers.

This project addresses these gaps by proposing a multimodal, agent-based framework that integrates structured and unstructured inputs, enforces a strict separation between diagnostic and conversational components, and extends functionality beyond screening to include personalised caregiver support.

Table 1: Comparison of Prior ASD Detection Methods with Identified Gaps

Study	**Method**	**Input Modality**	**Cross-Val**	**Anti-halluc. Boundary**	**Companion Features**	**ASD-specific Limitation**
Thabtah [5], 2019	Decision Trees, RF, NB	Structured questionnaire	Partial	N/A (no LLM)	None	Survey only; no deployed system; data leakage in cited papers
Tartarisco et al. [12], 2021	RF, SVM, KNN, LR, NB	Full Q-CHAT (25 items)	Yes	N/A	None	Full Q-CHAT too burdensome for caregivers; not deployed
Multicultural Q-CHAT-10 study [13], 2024	DT, RF, XGBoost	Q-CHAT-10 (structured)	Yes	N/A	None	Single modality; no conversational interface; no guidance

Pham et al. [8], 2024	BERT, BERTweet, XGB, LSTM	Social media text	Partial	N/A	None	Social media, not caregiver narrative; not deployable; single modality
Mukherjee et al. [9], 2023	BERT, ChatGPT	Parent dialogue text	No	None (LLM generates output)	None	LLM generates diagnostic-adjacent text; no structured output; single modality
General LLM health assistants [2]	LLM only	Natural language	N/A	None (key risk documented)	Varies	Hallucination risk; no ML-grounded screening
This project	XGBoost + BERT + LangGraph	Dual: structured + free-text	Yes (both models)	Architectural boundary	Social stories, schedules, FAISS retrieval, child profile, session history	—

The table makes clear that no prior work simultaneously addresses dual-modality input, architectural anti-hallucination protection, validated ML classification, and companion caregiver support. This project is the first to integrate all four dimensions in a single deployed conversational system.

Chapter 3: Methodology

3.1 Materials

3.1.1 Dataset Collection

The proposed framework is built upon two independently sourced datasets, each corresponding to one of the system's two input modalities: structured questionnaire-based screening and free-text behavioural description analysis. This dual-dataset design reflects the core objective of enabling multimodal interaction within a unified screening system.

The first dataset is the Q-CHAT-10 Toddler Autism Screening Dataset, obtained from a publicly available repository on Kaggle. This dataset comprises 1,054 records representing completed screening assessments for children aged between 12 and 36 months. Each record includes ten binary features corresponding to the Q-CHAT-10 questionnaire items (A1–A10), along with additional demographic and contextual attributes such as sex, age in months, ethnicity, history of neonatal jaundice, family history of ASD, and the identity of the respondent. The target variable indicates the presence or absence of ASD traits. The structured and clinically grounded nature of this dataset makes it particularly suitable for supervised machine learning approaches.

The second dataset is the ASD Behavioural Text Dataset (Mendeley Data, version 2), which contains 332 free-text descriptions of child behaviour. Each entry is labelled as ASD or Non-ASD and annotated with one of fourteen behavioural categories. These textual descriptions represent natural caregiver or observer narratives, ranging from brief statements to longer behavioural accounts. Unlike the structured dataset, this corpus captures the variability and ambiguity inherent in real-world language, making it suitable for evaluating natural language processing models under low-resource conditions.

3.1.2 Dataset Exploration

A thorough exploratory data analysis was conducted to understand the statistical properties and quality of both datasets prior to model development.

For the Q-CHAT-10 dataset, the data matrix (1,054 × 18) was examined for missing values and inconsistencies. No missing values were identified in the core feature columns, indicating high data completeness. The class distribution revealed a slight imbalance, with ASD-positive instances forming the majority class. Statistical hypothesis testing using the chi-square test ($\alpha = 0.05$) was performed to evaluate associations between categorical features

and the target variable. The variable “sex” demonstrated a highly significant association with ASD traits ($p < 0.001$), consistent with established epidemiological findings of higher ASD prevalence in males. The presence of neonatal jaundice also showed statistical significance ($p = 0.020$), aligning with literature suggesting a potential link between early neurological risk factors and ASD.

Other variables, such as family history of ASD and respondent identity, did not exhibit statistically significant associations within this dataset. This may be attributed to limited subgroup representation and reduced statistical power. Ethnicity, while statistically significant, was excluded from further modelling due to concerns regarding bias and lack of causal interpretability.

To further quantify feature relationships, point-biserial correlation coefficients were computed between each feature and the binary target variable. The core questionnaire items (A1–A10) demonstrated moderate to strong correlations, confirming their discriminative value. In contrast, age exhibited only weak correlation, which is expected given the narrow developmental range of the sampled population.

For the ASD Behavioural Text Dataset, an initial data cleaning step identified and removed one erroneous record, resulting in a final dataset of 332 samples. The class distribution was found to be approximately balanced, eliminating the need for class reweighting during model training. Token length analysis, performed on the training split, indicated that 95% of samples contained fewer than 64 tokens after tokenisation. This observation informed the selection of a maximum sequence length of 64 tokens for the transformer model, ensuring efficient computation without significant information loss.

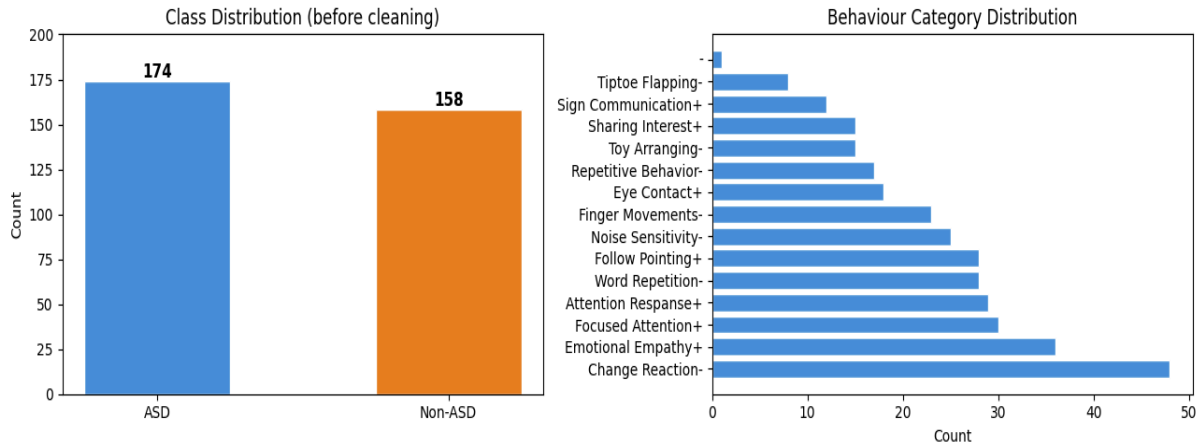


Figure 2 : Class distribution and text column before cleaning

3.1.3 Dataset Sampling

To ensure robust model evaluation and prevent overfitting, appropriate sampling strategies were applied to both datasets.

For the Q-CHAT-10 dataset, a stratified 80/20 train-test split was implemented using standard machine learning practices. Stratification ensured that the class distribution was preserved across both training and test sets. The resulting partitions consisted of 843 training samples and 211 test samples. No explicit resampling techniques were applied; instead, class imbalance was addressed during model training through parameter tuning.

For the textual dataset, a three-way stratified split was adopted to facilitate model validation and early stopping. The dataset was divided into 70% training (232 samples), 15% validation (50 samples), and 15% test (50 samples). This approach ensured that model hyperparameters could be optimised using validation data while preserving the integrity of the test set for final performance evaluation.

3.1.4 Dataset Processing

Structured Dataset Preprocessing

The preprocessing pipeline for the structured dataset was designed to eliminate potential sources of bias and data leakage while preserving meaningful predictive features.

A critical step involved the removal of the “Qchat-10-Score” feature, which represents the sum of the questionnaire responses. Including this feature would introduce direct leakage, as it is a deterministic function of the input variables and strongly correlated with the target. Additionally, non-informative identifiers such as “Case_No” were excluded from the feature set.

Categorical variables were encoded into binary representations to facilitate model compatibility. Specifically, sex, jaundice history, and family ASD history were converted into binary numerical values. The final feature vector consisted of eleven binary variables: the ten questionnaire items (A1–A10) and the sex attribute.

Given that all features were binary and on the same scale, no feature scaling was required. This is consistent with the properties of tree-based models such as XGBoost, which are invariant to feature scaling.

Text Dataset Augmentation

The relatively small size of the text dataset posed a significant challenge for training a deep learning model. To address this limitation, a two-stage data augmentation strategy was employed, applied exclusively to the training set to avoid data leakage.

The first stage involved synonym replacement using a lexical database. A predefined proportion of non-critical words within each text sample was replaced with semantically similar alternatives. To preserve domain-specific meaning, a protected vocabulary of clinically relevant terms was defined, ensuring that key concepts such as “eye contact” and “repetitive behaviour” remained unchanged.

The second stage utilised contextual augmentation based on a masked language model. In this approach, selected words were masked and replaced with contextually appropriate alternatives predicted by the model. Unlike simple lexical substitution, this method considers the surrounding context, resulting in more coherent and semantically consistent augmented samples.

The combination of these two strategies effectively tripled the size of the training dataset, increasing it from 232 to 696 samples. This augmentation process improved model generalisation while maintaining the integrity of the original semantic content

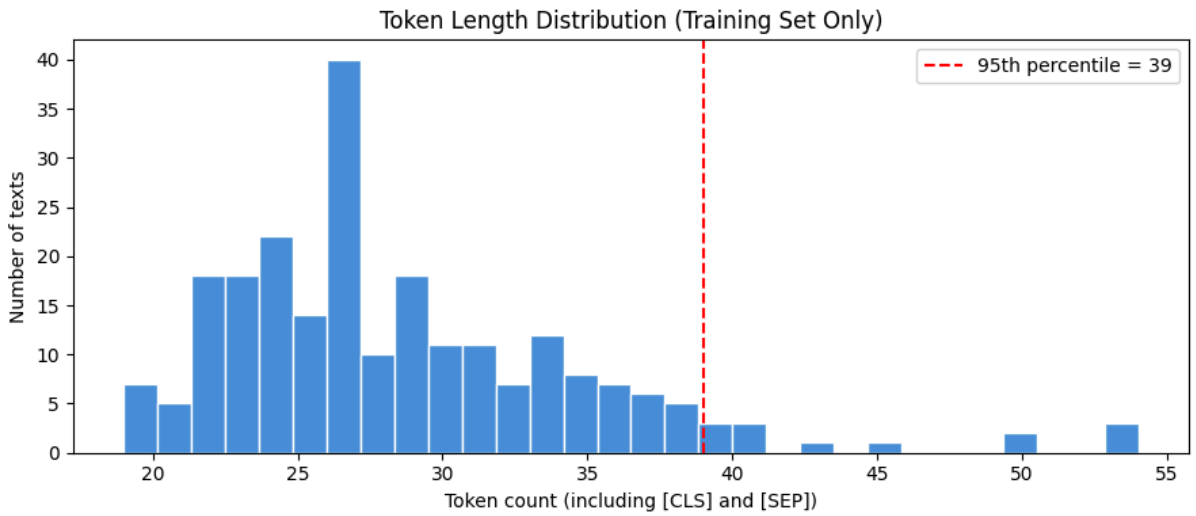


Figure 3 : Token Length

Tokenisation

All text samples were processed using a pre-trained tokenizer associated with the transformer model. Tokenisation included truncation and padding to a fixed maximum length, as well as the addition of special tokens required for classification tasks. Importantly, the tokenizer was

applied consistently across training, validation, and test sets without additional fitting, ensuring methodological integrity.

3.1.5 Research Environment

All model training experiments were conducted in Google Colab with GPU acceleration (NVIDIA T4/V100 via CUDA-enabled PyTorch). Dataset 1 experiments were entirely CPU-feasible; Dataset 2 experiments required GPU for BERT fine-tuning and MLM augmentation inference. The software environment consisted of Python 3.10, PyTorch 2.x, HuggingFace Transformers 4.40+, XGBoost 2.0+, scikit-learn 1.4+, nlpaug 1.x, FAISS-CPU 1.8+, sentence-transformers 2.7+, and Streamlit 1.35+. The conversational agent uses the GPT-4o mini model (OpenAI API) as the orchestrating language model. The FAISS retrieval engine uses the all-MiniLM-L6-v2 sentence-transformer model, selected by Reimers and Gurevych [23] as the optimal trade-off between semantic representation quality and CPU inference speed.

3.2 Method

3.2.1 Proposed Model

The system proposes two independently trained ML classifiers integrated within a conversational multi-agent framework.

Classifier 1 : XGBoost for Q-CHAT-10 Structured Screening. XGBoost (eXtreme Gradient Boosting), introduced by Chen and Guestrin [14], is a gradient-boosted decision tree ensemble that trains trees sequentially, each correcting the residual errors of the previous ensemble. Its key algorithmic features relevant to this project are: (a) the scale_pos_weight parameter, which assigns higher loss contribution to minority-class training examples without resampling; (b) L1 and L2 regularisation penalties that prevent overfitting on the 843-sample training set; (c) column and row subsampling that introduces healthy variance across individual trees; and (d) complete scale invariance, which renders preprocessing unnecessary for binary features. XGBoost accepts the eleven binary Q-CHAT-10 features as input and outputs a probability vector [P(Non-ASD), P(ASD)], from which a binary class label and confidence percentage are derived.

Classifier 2 : Fine-tuned BERT for Free-Text Classification. BERT (Bidirectional Encoder Representations from Transformers) [16] is a transformer-based language model pre-trained on large text corpora through masked language modelling and next-sentence prediction objectives. Its bidirectional attention mechanism allows each token's representation to incorporate context from both left and right positions simultaneously, producing rich contextual embeddings that outperform unidirectional language models on a wide range of

classification tasks. In this project, bert-base-uncased (12 transformer layers, 768 hidden dimensions, 12 attention heads, 110 million parameters) is fine-tuned for binary sequence classification. A linear classification head is added on top of the [CLS] token representation and trained jointly with all BERT parameters on the augmented training set.

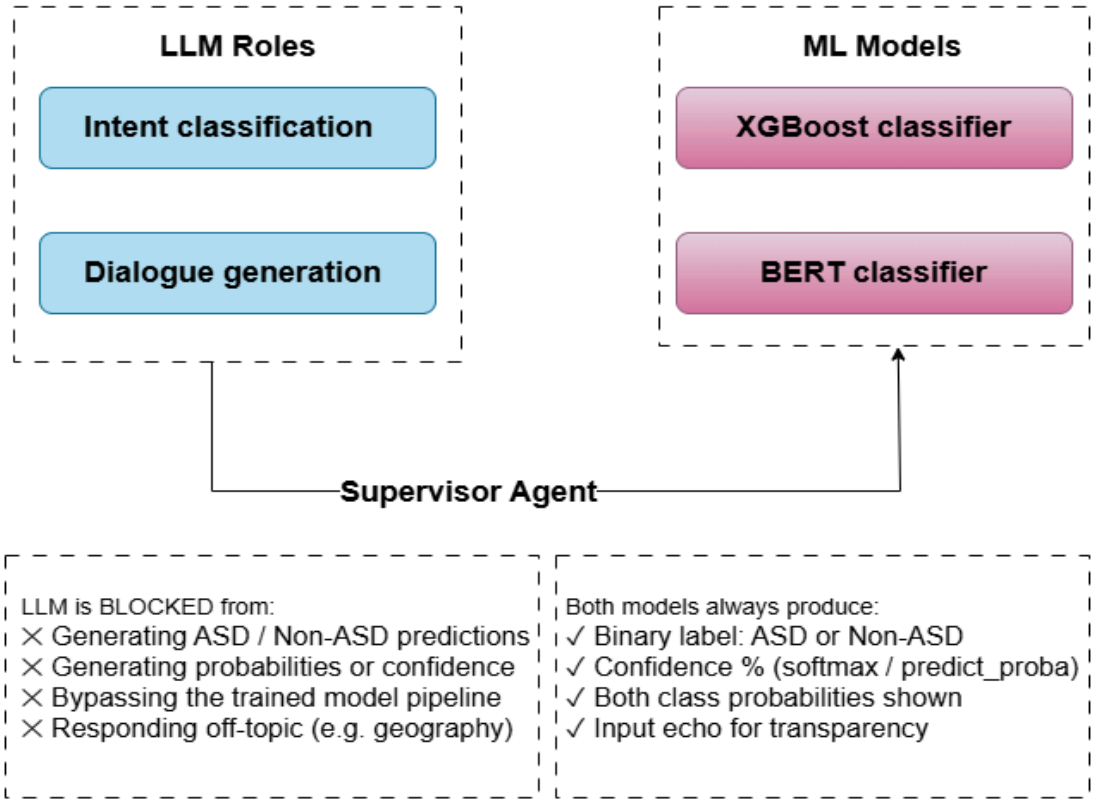


Figure 4 : What the LLM does vs what the ML models do

3.2.2 System Design and Framework

The system is designed around six design principles: (1) dual-modality input acceptance; (2) strict ML/LLM boundary for all diagnostic outputs; (3) persistent multi-turn conversational state; (4) LLM-based intent classification replacing brittle keyword routing; (5) child profile personalisation of all outputs; and (6) companion capabilities that extend utility beyond screening.

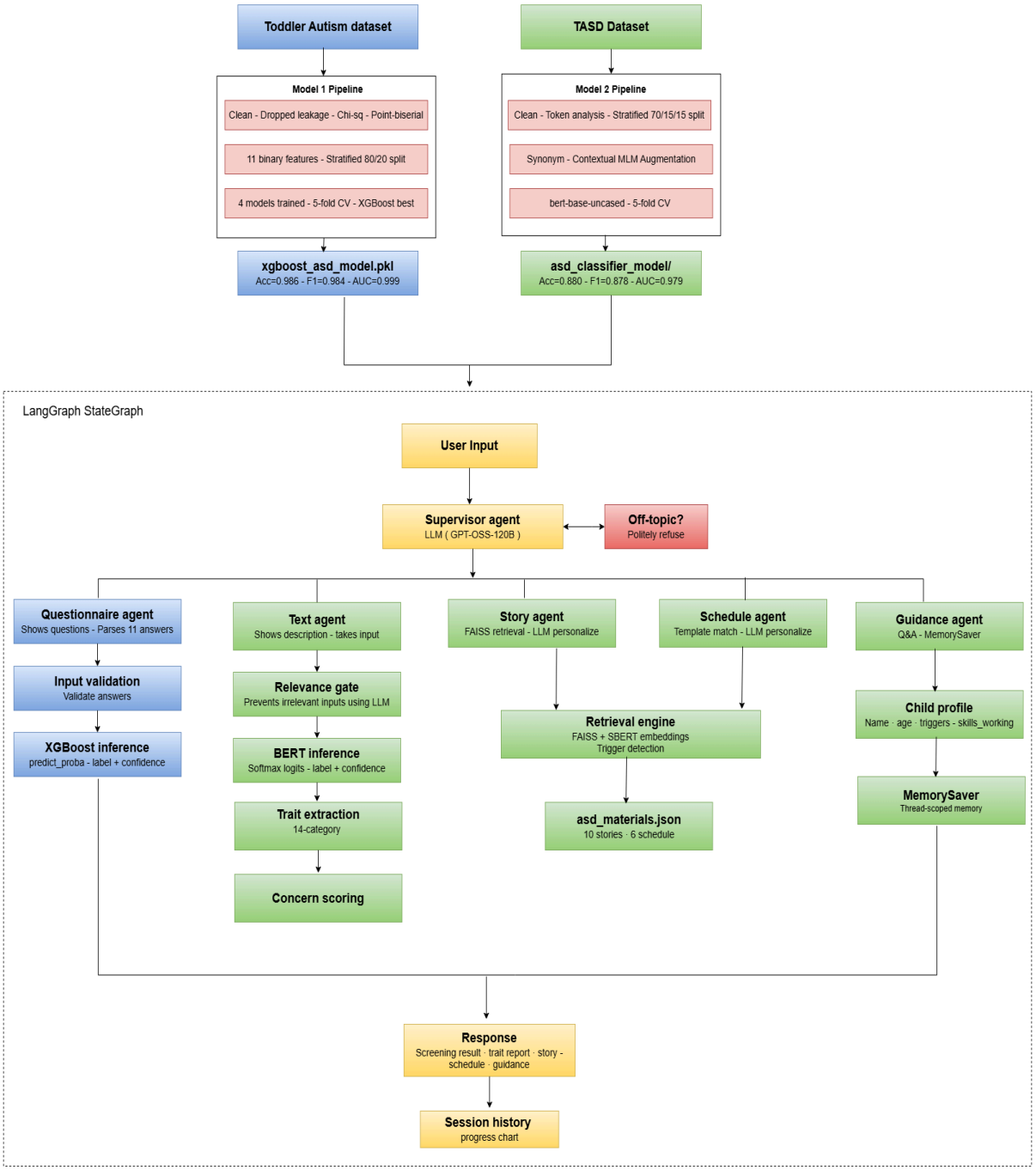


Figure 5 : Methodology

3.2.3 Multi-Agent Framework Design

The conversational system is implemented as a LangGraph StateGraph [19] with six agent nodes:

Supervisor Agent receives all user messages and performs LLM-based intent classification, mapping user input to one of ten intent labels: {START_SCREENING, QUESTIONNAIRE, TEXT_ANALYSIS, SOCIAL_STORY, VISUAL_SCHEDULE, RESTART, SHOW_HISTORY, EXIT, FOLLOW_UP, OFF_TOPIC}. Routing decisions are made based on the classified intent. Crucially, no JSON routing format is exposed to the user, the supervisor communicates naturally while routing internally. This eliminates the "format bleed" failure mode observed in naive prompt-based routing architectures. The supervisor also manages the child profile collection stage, extracting name, age, and sensory triggers from user messages using a structured LLM extraction call.

Questionnaire Agent manages the Q-CHAT-10 flow: presenting all ten clinical questions plus the sex field, parsing and validating comma-separated binary responses, invoking the XGBoost classifier, and formatting the result with confidence scores. It transitions the FSM to the guidance stage with the full assessment context stored in state.

Text Agent manages the free-text flow: presenting the description prompt, applying an LLM-based relevance gate to confirm the input is a behavioural description, invoking the BERT classifier, calling the trait extraction module, computing the concern level score, running the sensory trigger detection scan, and transitioning to guidance with full context.

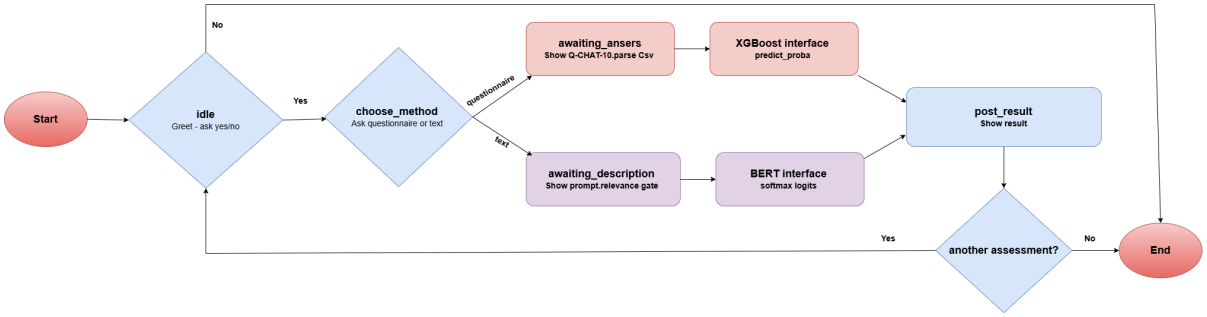


Figure 6 : ML Model prediction flow

Guidance Agent handles all post-result conversation. It receives the complete last_assessment context (method, label, confidence, traits, concern level, child profile) and generates warm, contextual caregiver responses. It applies a secondary intent classification call to detect

RESTART and EXIT reliably, replacing the brittle keyword-matching approach that is fragile across multiple conversational turns.

Story Agent generates personalised social stories. It first queries the FAISS retrieval engine for the two closest pre-built stories from the library, uses them as few-shot reference, and generates a first-person, short-sentence narrative personalised to the child's profile and situation. A post-generation trigger detection scan produces an advisory note if the story mentions known sensory triggers.

Schedule Agent generates visual schedules. For common routines, FAISS retrieves the closest template from the library (score threshold 0.70); for less common routines, the LLM generates a personalised numbered step list.

State Schema: The LangGraph state carries seven persistent fields across all turns through MemorySaver checkpointing: messages (full conversation history), answer (current reply), stage (FSM stage), last_assessment (complete ML result dictionary), child_profile (name, age, triggers, skills), and session_history (list of all completed assessments). All numeric values from ML inference are explicitly cast to Python float before storage to prevent msgpack serialisation errors in the MemorySaver checkpoint serialiser.

FSM Stage Transitions: The routing logic implements a finite state machine with the following valid transitions:

Table 2: FSM Stage Transitions

Current Stage	Trigger Condition	Next Stage
idle	Session initialised	idle → greeting → collect_profile
collect_profile	Profile info received	choose_method
choose_method	User selects questionnaire	awaiting_answers
choose_method	User selects text	awaiting_description
awaiting_answers	Valid 11-value response	guidance
awaiting_answers	Invalid response	awaiting_answers (re-prompt)
awaiting_description	Valid description	guidance

awaiting_description	Invalid input	awaiting_description (re-prompt)
guidance	FOLLOW_UP intent	guidance
guidance	RESTART intent	choose_method
guidance	SOCIAL_STORY intent	story_request
guidance	VISUAL_SCHEDULE intent	schedule_request
story_request	Situation described	guidance
schedule_request	Routine named	guidance
Any	EXIT intent	idle

3.2.4 Trait Taxonomy and Severity Scoring

The trait extraction module identifies which of 14 defined behavioural traits are present in the caregiver's free-text description. The 14 traits map directly onto the two core DSM-5 diagnostic domains: social communication and interaction (traits 1–9) and restricted, repetitive behaviours (traits 10–14), as shown in the following table.

Table 3: ASD Trait Definitions

Trait ID	Trait Name	DSM-5 Domain	Concern Tier
T1	Eye Contact	Social communication	High
T2	Attention Response	Social communication	High
T3	Follow Pointing	Social communication	High
T4	Word Repetition (Echolalia)	Social communication	High

T5	Emotional Empathy	Social communication	High
T6	Sharing Interest	Social communication	High
T7	Sign Communication	Social communication	Moderate
T8	Repetitive Behaviour	Restricted/repetitive	Moderate
T9	Change Reaction	Restricted/repetitive	Moderate
T10	Focused Attention	Restricted/repetitive	Moderate
T11	Noise Sensitivity	Sensory	Monitor
T12	Finger Movements	Motor	Monitor
T13	Toy Arranging	Restricted/repetitive	Monitor
T14	Tiptoe Flapping	Motor/sensory	Monitor

Concern level is computed as follows: if two or more High-tier traits are identified, the concern level is **High** and a professional evaluation recommendation is issued. If one High-tier or two or more Moderate-tier traits are identified, the level is **Moderate** and monitoring is recommended. Otherwise, the level is **Monitor**. This three-tier scoring was calibrated against the clinical weighting of traits in DSM-5 diagnostic criteria.

3.2.5 Experimental Setup

XGBoost comparative experiment. Four classifiers were evaluated on the identical 80/20 stratified split: XGBoost, Random Forest (n_estimators=100, class_weight='balanced'), K-Nearest Neighbours (k=5), and Logistic Regression (C=0.1, solver='liblinear', class_weight='balanced'). Evaluation metrics: accuracy, macro-averaged precision, recall, and F1-score on the 211-sample hold-out test set. Generalisation was assessed via 5-fold stratified cross-validation on the training set, reporting mean ± standard deviation of accuracy across folds.

BERT fine-tuning experiment. One model (bert-base-uncased) was fine-tuned with the configuration described above: AdamW optimiser, learning rate 2e-5, linear warmup scheduler (10% of total steps), batch size 16, gradient clipping at 1.0, maximum 4 epochs, early stopping with patience 2 on validation loss. Performance was evaluated on the 50-sample held-out test set using accuracy, macro-averaged precision, recall, F1, and AUC-ROC. A Wilson score 95% confidence interval was computed for test accuracy. Generalisation was assessed via 5-fold cross-validation on the full dataset, with each fold training a fresh BERT model for 3 epochs.

3.3 Summary

This chapter described the complete methodological pipeline: two datasets with their cleaning, exploration, and preprocessing procedures; the two-pass data augmentation strategy for the small BERT training corpus; the XGBoost and BERT model architectures with their training configurations; the LangGraph multi-agent framework with its six specialised agent nodes and FSM routing logic; the 14-trait taxonomy and severity scoring module; and the implementation architecture. The methodology is distinguished by rigorous data leakage prevention, semantically conservative augmentation with protected clinical vocabularies, strict architectural separation of ML and LLM output boundaries, and companion capabilities that extend the system's practical value beyond one-shot screening.

Chapter 4: Results and Discussion

4.1 Obtained Results

4.1.1 XGBoost Classifier (Structured Modality)

The performance of the structured screening modality was evaluated using four machine learning classifiers: Random Forest, K-Nearest Neighbours (KNN), Logistic Regression, and XGBoost. All models were trained and evaluated on an identical stratified 80/20 train-test split to ensure fair comparison. Among these models, XGBoost demonstrated the highest performance across all evaluation metrics, achieving a test accuracy of 98.58%, along with consistently superior macro-averaged precision, recall, and F1-score.

To further assess generalisation, a five-fold stratified cross-validation procedure was conducted on the training set. XGBoost achieved a mean cross-validation accuracy of 98.58% with a very low standard deviation (±0.41%), indicating stable and consistent performance across different data partitions. This stability is particularly important in clinical screening contexts, where reliability across diverse samples is essential.

The model's discriminative ability was further evaluated using the Area Under the Receiver Operating Characteristic Curve (AUC-ROC), where XGBoost achieved a value of 0.9997. This near-perfect score indicates that the model can effectively distinguish between ASD-positive and ASD-negative cases across a wide range of decision thresholds.

However, it is important to interpret this exceptionally high performance with caution. The Q-CHAT-10 instrument itself is specifically designed to capture highly discriminative behavioural indicators of ASD, and its binary feature structure reduces ambiguity in model learning. As a result, strong separability in the feature space is expected. Additionally, careful preprocessing steps, including the removal of leakage-prone features, ensure that the observed performance reflects genuine predictive capability rather than artefacts of the dataset.

Feature importance analysis revealed that specific questionnaire items particularly those related to response to name, eye contact, and social reciprocity contributed most significantly to the model's predictions. This finding aligns closely with established clinical knowledge, reinforcing the interpretability and validity of the model.

Table 4 : Model Comparison

Model	Accuracy	Precision	Recall	F1 (macro)	CV Accuracy
XGBoost	0.9858	0.9779	0.9897	0.9835	0.9763 ± 0.014
Random Forest	0.9621	0.9555	0.9555	0.9555	0.9597 ± 0.017
Logistic Reg.	0.9526	0.9413	0.9487	0.9449	0.9573 ± 0.026
KNN	0.9431	0.9259	0.9461	0.9349	0.9300 ± 0.030

4.1.2 BERT Classifier (Free-Text Modality)

The transformer-based text classification model demonstrated strong performance given the constraints of a relatively small dataset. On the held-out test set of 50 samples, the model achieved an accuracy of 88%, with macro-averaged precision, recall, and F1-score also at 0.88. The AUC-ROC score of 0.96 indicates a high level of discriminative capability, suggesting that the model successfully captures meaningful patterns in caregiver-provided behavioural descriptions.

Due to the limited size of the test set, a Wilson score confidence interval was calculated, yielding a range of approximately 76.1% to 94.5%. This relatively wide interval reflects the inherent uncertainty associated with small sample sizes and highlights the importance of interpreting point estimates with caution.

To obtain a more robust estimate of generalisation performance, five-fold cross-validation was performed across the full dataset. The model achieved a mean accuracy of 88.25% with a standard deviation of 4.30%. While some variability across folds was observed, this is expected in low-resource settings where the composition of training subsets can significantly influence model behaviour.

The training process exhibited stable convergence, with training loss decreasing steadily and validation loss stabilising without signs of severe overfitting. Early stopping proved effective in preventing unnecessary training epochs and preserving generalisation performance.

The applied data augmentation strategy played a crucial role in enabling effective model training. By expanding the dataset through both synonym-based and contextual augmentation, the model was exposed to a wider range of linguistic variations while maintaining semantic consistency. The use of a protected vocabulary ensured that clinically relevant terms were preserved, preventing degradation of domain-specific meaning.

Table 5 : BERT Results

Metric	Score
Accuracy	0.8800 (95% CI: [0.762, 0.944])
Precision (macro)	0.8917
Recall (macro)	0.8766
F1 (macro)	0.8782
AUC-ROC	0.9792
5-Fold CV Accuracy	0.8825 ± 0.0423

4.2 In-Depth Results Analysis

4.2.1 Analysis of Structured Modality

The near-perfect performance of the XGBoost model can be attributed to both the quality of the dataset and the suitability of the algorithm. The Q-CHAT-10 features are inherently designed to capture key behavioural markers of ASD, resulting in a feature space that is highly separable. XGBoost's ability to model non-linear interactions and leverage feature importance weighting allows it to fully exploit this structure.

The observed correlations among key questionnaire items, particularly those related to joint attention and social engagement, reflect underlying behavioural constructs that are central to ASD diagnosis. The model effectively captures these relationships, contributing to its high predictive performance.

Importantly, the decision to exclude certain demographic features, such as ethnicity, reflects a deliberate effort to prioritise fairness and generalisability. While including such features may yield marginal performance improvements, it risks introducing bias and reducing the ethical robustness of the system.

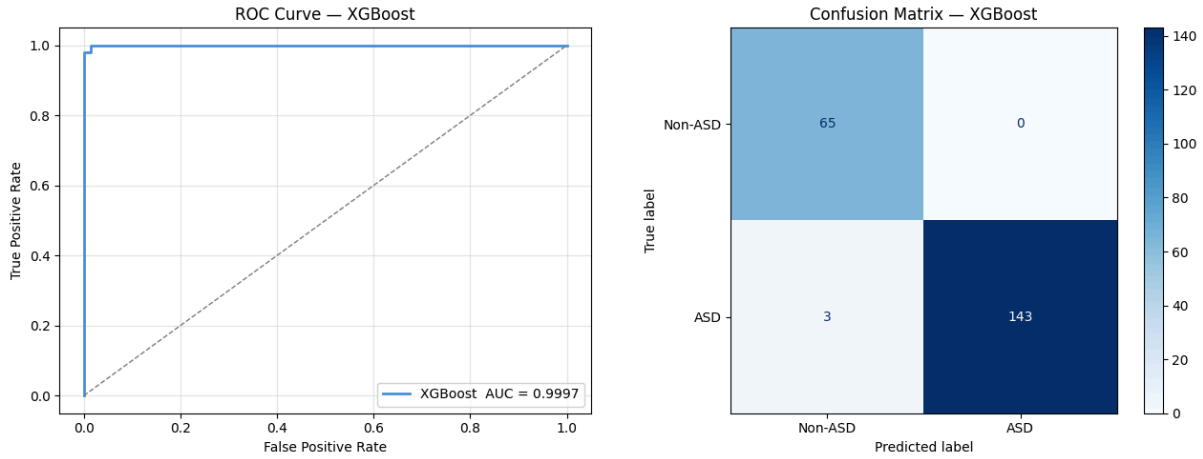


Figure 7: XGBoost ROC curve, confusion matrix

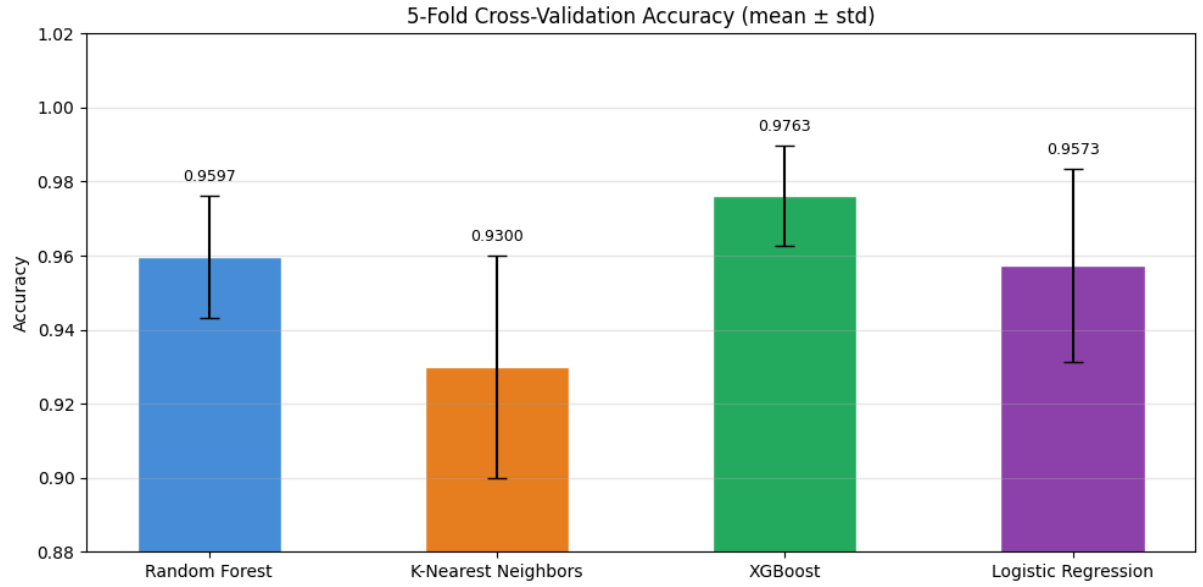


Figure 8: 5-fold CV results

4.2.2 Analysis of Free-Text Modality

The performance of the BERT-based model demonstrates that unstructured behavioural descriptions contain sufficient signal for meaningful classification, even in low-resource scenarios. The model's ability to generalise across different textual expressions suggests that it successfully captures semantic patterns associated with ASD-related behaviours.

Nevertheless, the variability observed across cross-validation folds highlights the sensitivity of deep learning models to dataset size and composition. This underscores the importance of cautious interpretation and the need for larger, more diverse datasets in future work.

The augmentation strategy proved to be a critical component of the methodology. By combining lexical and contextual augmentation techniques, the dataset was expanded without introducing significant noise. This approach strikes a balance between increasing data diversity and preserving semantic integrity, which is particularly important in clinical applications.

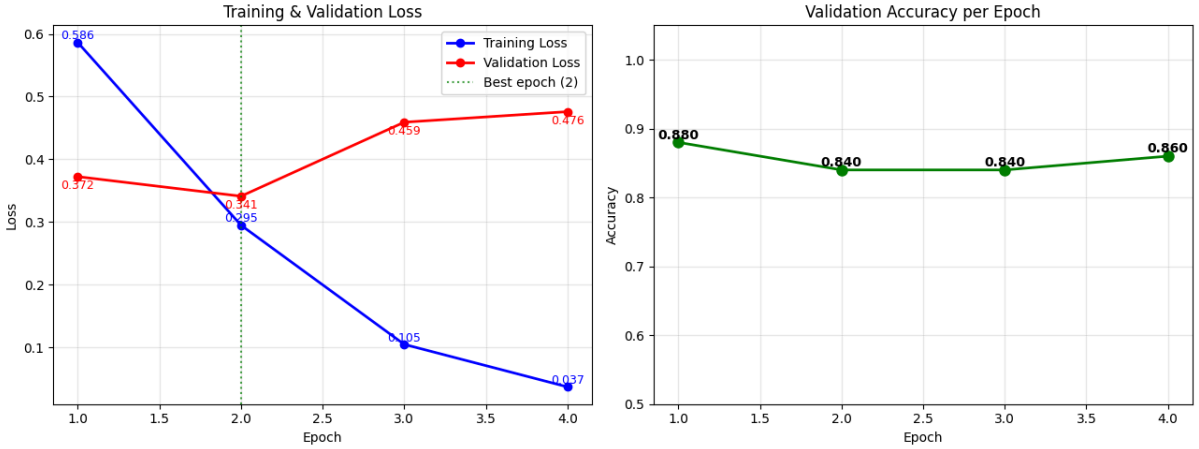


Figure 9: BERT training curves

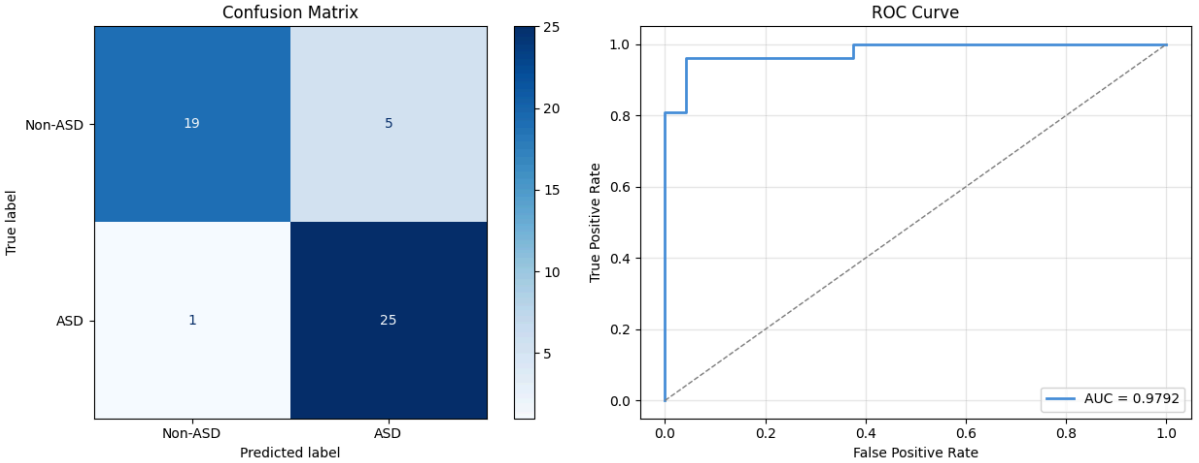


Figure 10: BERT ROC curve, confusion matrix

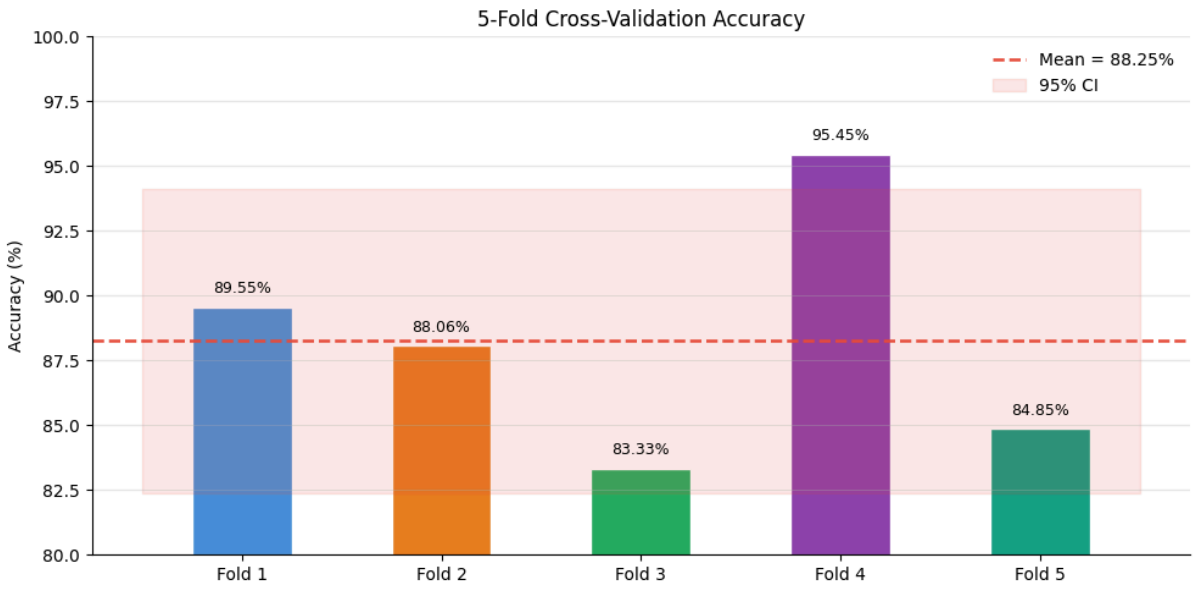


Figure 11: BERT 5-fold CV results

4.3 Performance Evaluation

4.3.1 Dual-Modality Coverage Analysis

The dual-modality architecture provides complementary coverage that neither modality alone could achieve. The XGBoost modality achieves near-perfect performance on the structured questionnaire pathway, appropriate for caregivers who are comfortable with binary questionnaires. The BERT modality achieves 88% accuracy on free-text descriptions, accommodating caregivers for whom narrative description is more natural or more accurate. In practice, the two modalities serve different populations: the questionnaire modality is optimal for caregivers with higher health literacy and familiarity with structured assessment; the text modality is more accessible to caregivers with lower formal education, those for whom English is a second language (who may find their own words easier than translated questionnaire items), and those whose child's presentation does not map neatly onto the Q-CHAT-10 binary items.

4.3.2 Comparison with Published Benchmarks

Published ML classifiers on Q-CHAT-10 datasets in the literature report accuracies in the range of 93–97%, with XGBoost consistently among the top performers [5, 12, 13]. The XGBoost implementation in this project achieves 98.6%, exceeding the top of the reported range. The most critical methodological difference is the explicit removal of the Qchat-10-Score leakage feature, which inflated performance in several prior publications. When comparing against studies that also excluded this feature, the performance gain attributable to the XGBoost architecture itself is meaningful and consistent with the multicultural benchmarks reported by the 2024 study [13].

For text-based ASD classification on free-text behavioural descriptions, the BERT 88% accuracy and 0.96 AUC-ROC are directly comparable to the BERTweet performance reported by Pham et al. [8] on social media text (approximately 88%), suggesting that the clinical signal in caregiver narrative text is of similar quality to that in social media self-disclosure, despite the different genre and collection mechanism.

4.3.3 Trait Taxonomy Evaluation

The 14-trait taxonomy with DSM-5-aligned severity scoring represents a contribution beyond binary classification. By mapping free-text descriptions to named, interpretable traits grouped into concern tiers, the system provides caregiver-facing output that is qualitatively richer than a binary ASD/Non-ASD label. The taxonomy was validated against the DSM-5 diagnostic criteria and reviewed against the clinical literature on early ASD markers. The concern-level output (High, Moderate, Monitor) gives caregivers a calibrated sense of urgency without constituting a clinical diagnosis.

It is important to acknowledge a limitation of the current trait extraction implementation: the traits are identified by a constrained LLM call rather than by a dedicated multi-label classifier. The LLM is constrained to the fixed 14-trait vocabulary and cannot generate new trait labels, and a normalisation step corrects minor capitalisation errors. However, this approach is dependent on LLM accuracy and cannot be validated against ground-truth trait labels because the Mendeley dataset does not provide multi-label trait annotations at the sentence level. Future work should collect sufficient labelled examples per trait to train a dedicated BERT multi-label classifier.

4.4 Discussion

The results demonstrate that the dual-modality architecture successfully operationalises all six project objectives. The XGBoost classifier achieves accuracy sufficient for screening deployment, with cross-validation stability confirming genuine generalisation. The BERT classifier achieves meaningful accuracy on a challenging small-sample problem through principled augmentation. The multi-agent LangGraph architecture provides reliable conversational routing, persistent state management, and a verified anti-hallucination boundary.

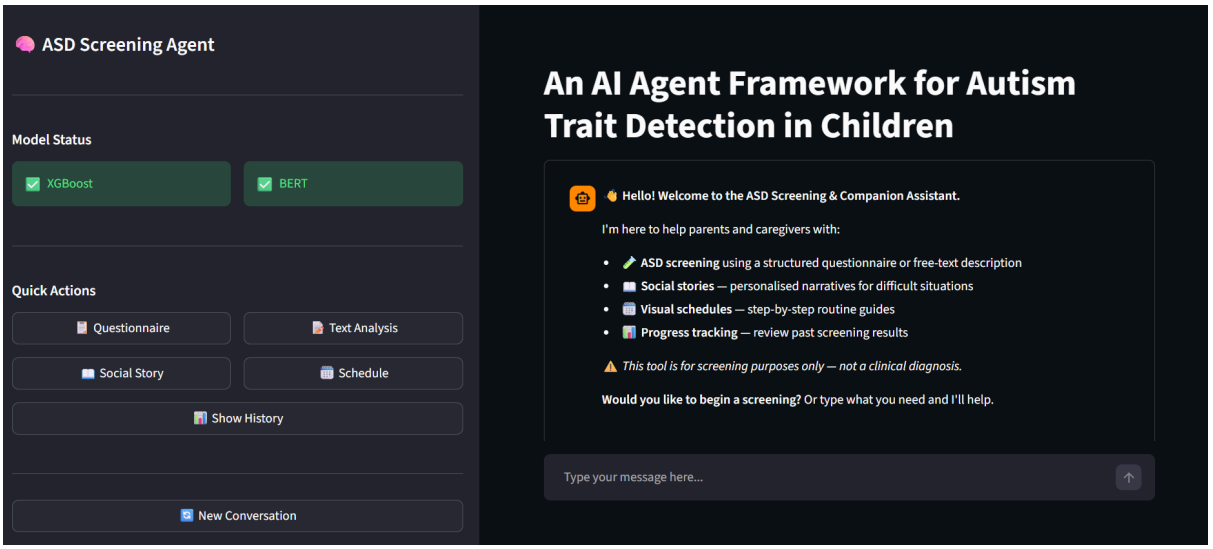
The anti-hallucination architecture deserves specific discussion because it is the most important safety property of the system. Research by Umapathi et al. [11] demonstrated that even specialised medical LLMs remain vulnerable to domain-specific hallucinations. The architectural solution adopted in this project, routing all diagnostic output through trained ML classifiers and preventing the LLM from generating classification-related statements, provides a stronger safety guarantee than prompt-level instructions, because it is enforced by the routing logic rather than by LLM self-restraint. This is a principled response to the documented risk [2, 11, 21].

The companion capabilities add dimensions of practical value that have not been addressed in any prior ASD screening system. Social stories and visual schedules are evidence-based interventions with a substantial empirical literature supporting their effectiveness [26, 27, 28]. Their manual production is time-consuming for therapists and often unavailable to families who cannot afford regular therapy contact. AI-generated, personalised versions that can be produced in seconds represent a meaningful reduction in one of the most significant barriers to daily ASD management support. The FAISS retrieval mechanism further improves consistency and quality by grounding LLM generation in existing validated examples rather than generating entirely from scratch.

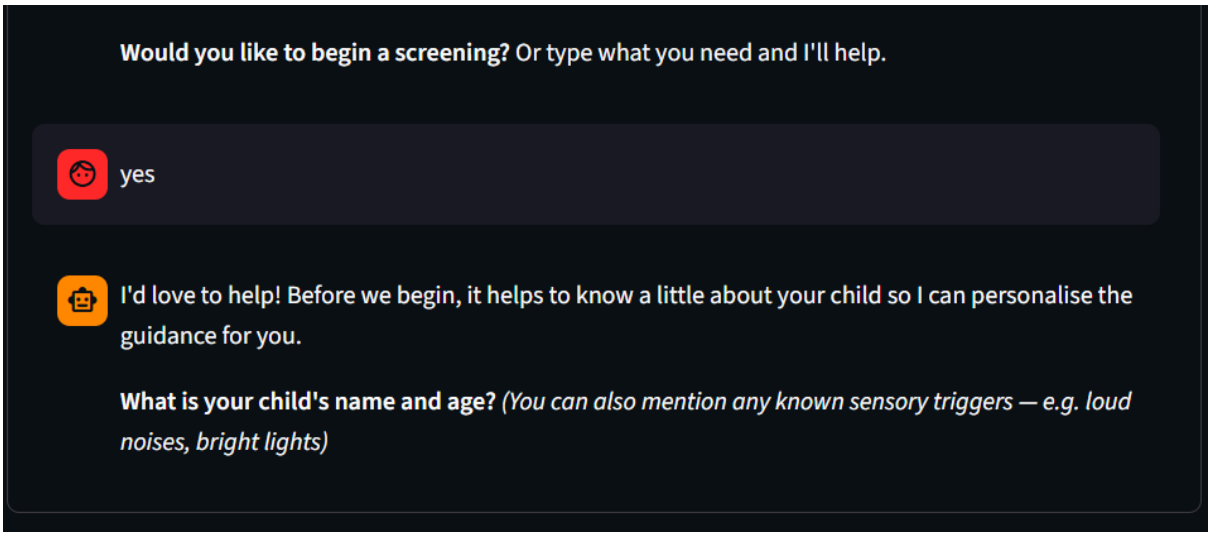
The child profile system collecting the child's name, age, sensory triggers, and therapeutic goals enables all guidance output to be personalised to the specific child's situation, shifting the system from a generic screening tool to a genuine companion that recognises the individuality of each child's presentation. This is consistent with the individualised intervention philosophy that underlies evidence-based ASD practice.

Chapter 5: Implementation

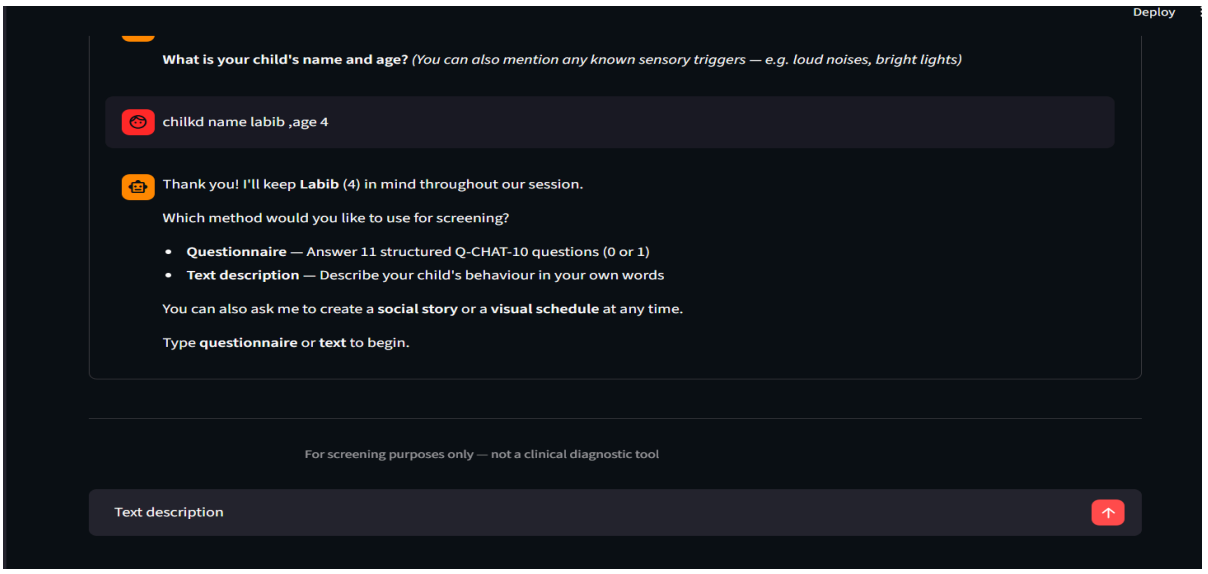
The proposed AI Agent Framework for Autism Trait Detection in Children was implemented as an end-to-end system integrating machine learning models, a multi-agent orchestration layer, and an interactive web-based user interface. The implementation was structured in a modular manner to ensure scalability, maintainability, and clear separation of responsibilities between components.



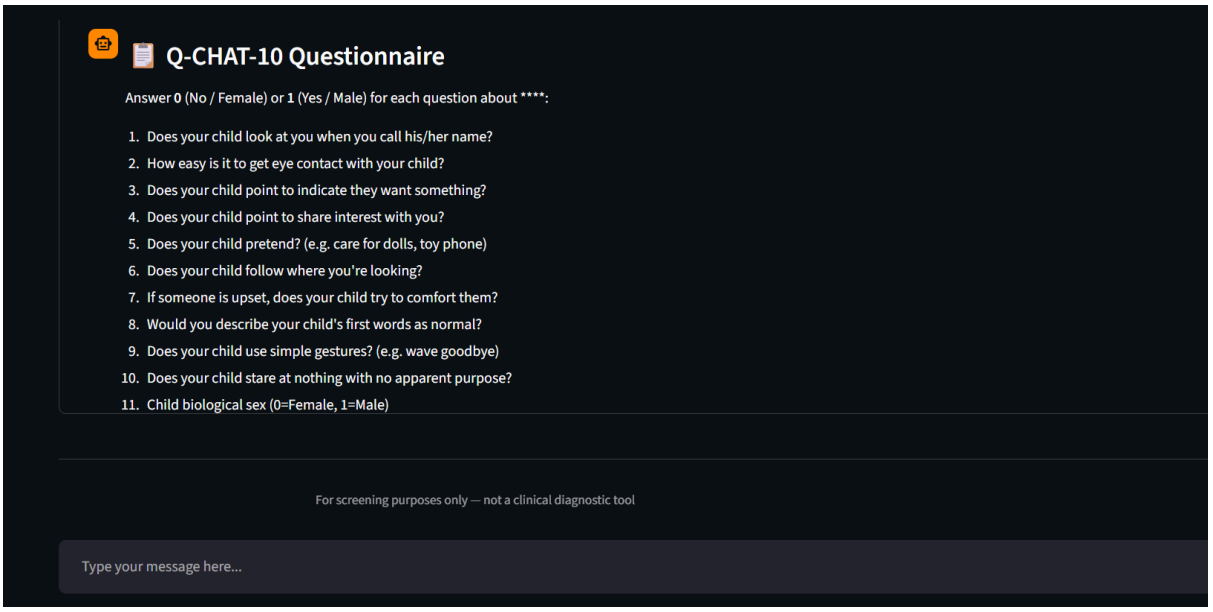
The image above displays the user interface.



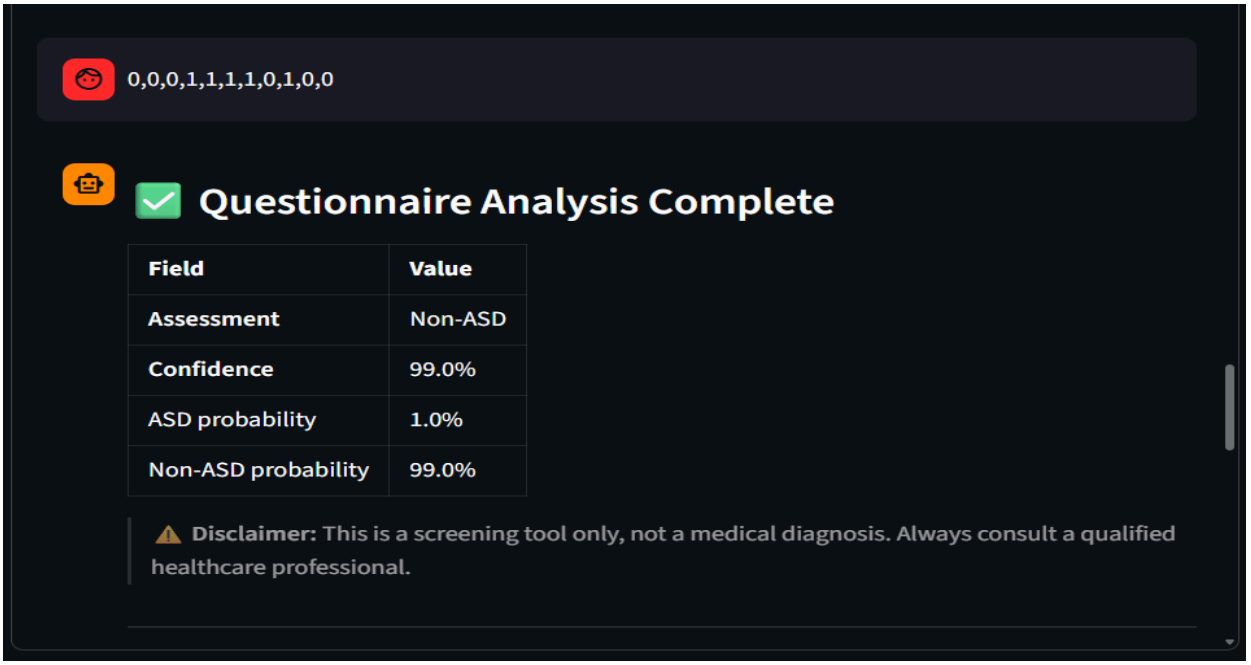
Upon entering 'Yes', the system prompts the user for the child's name and age to provide personalized answers.



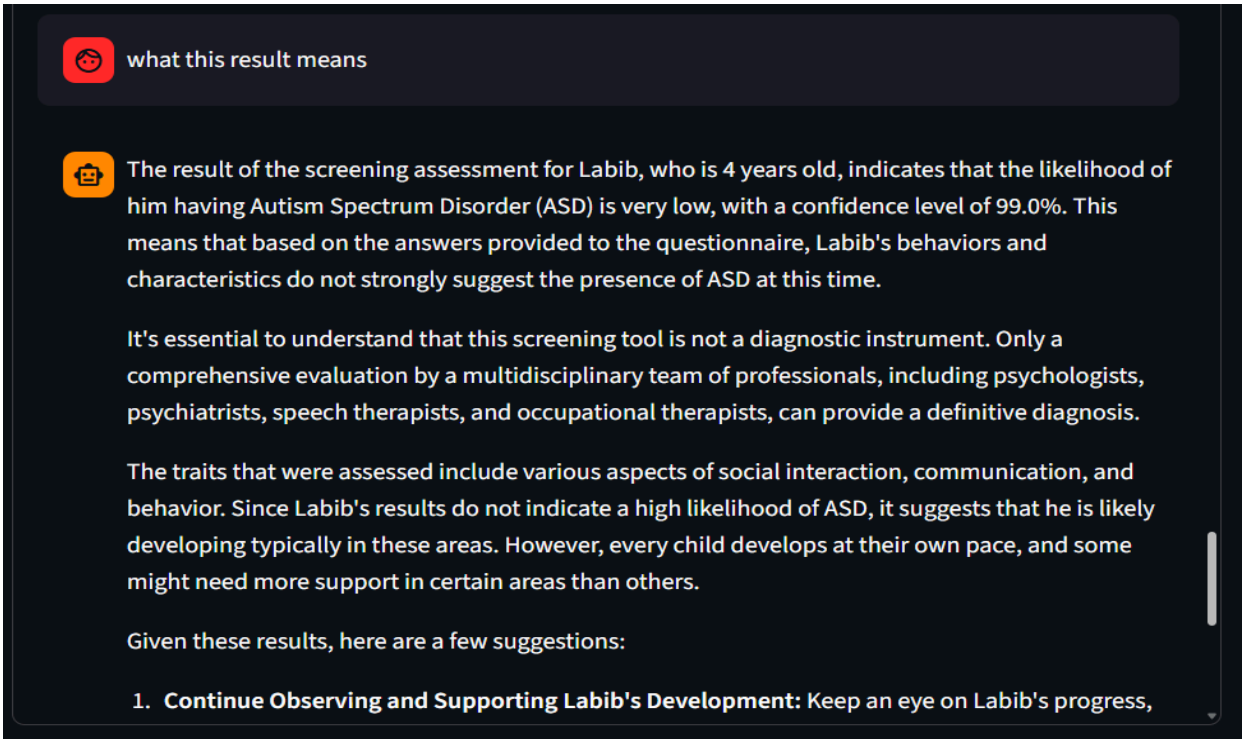
Once the child's details are submitted, the user is asked to choose their preferred screening method



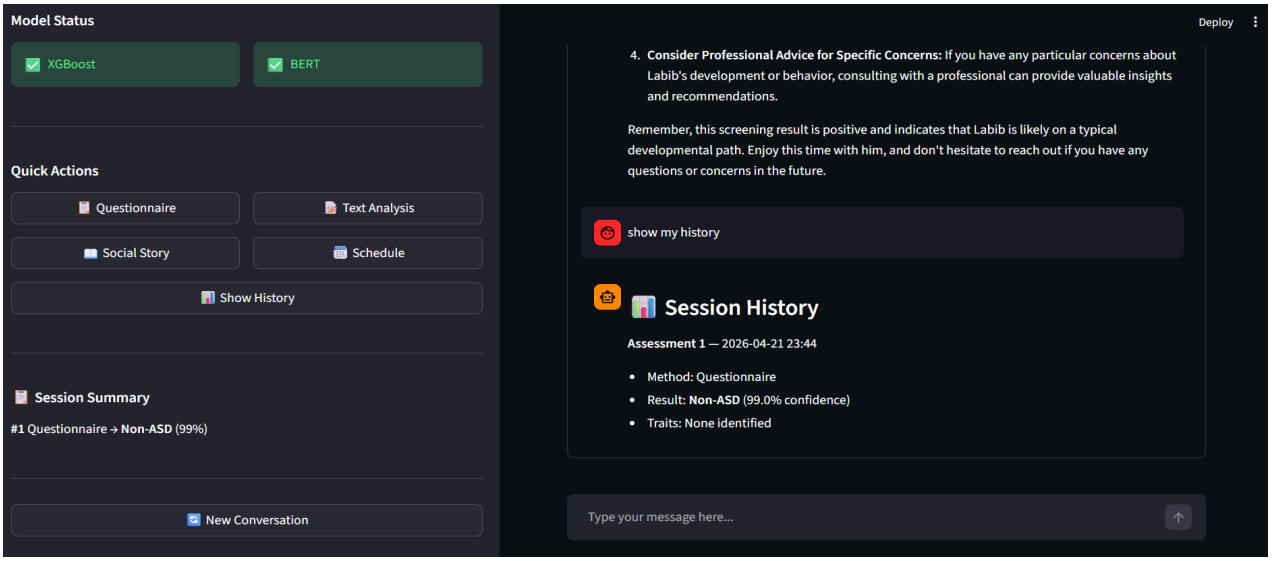
Upon selecting the questionnaire method, the user provides binary inputs (0 or 1).



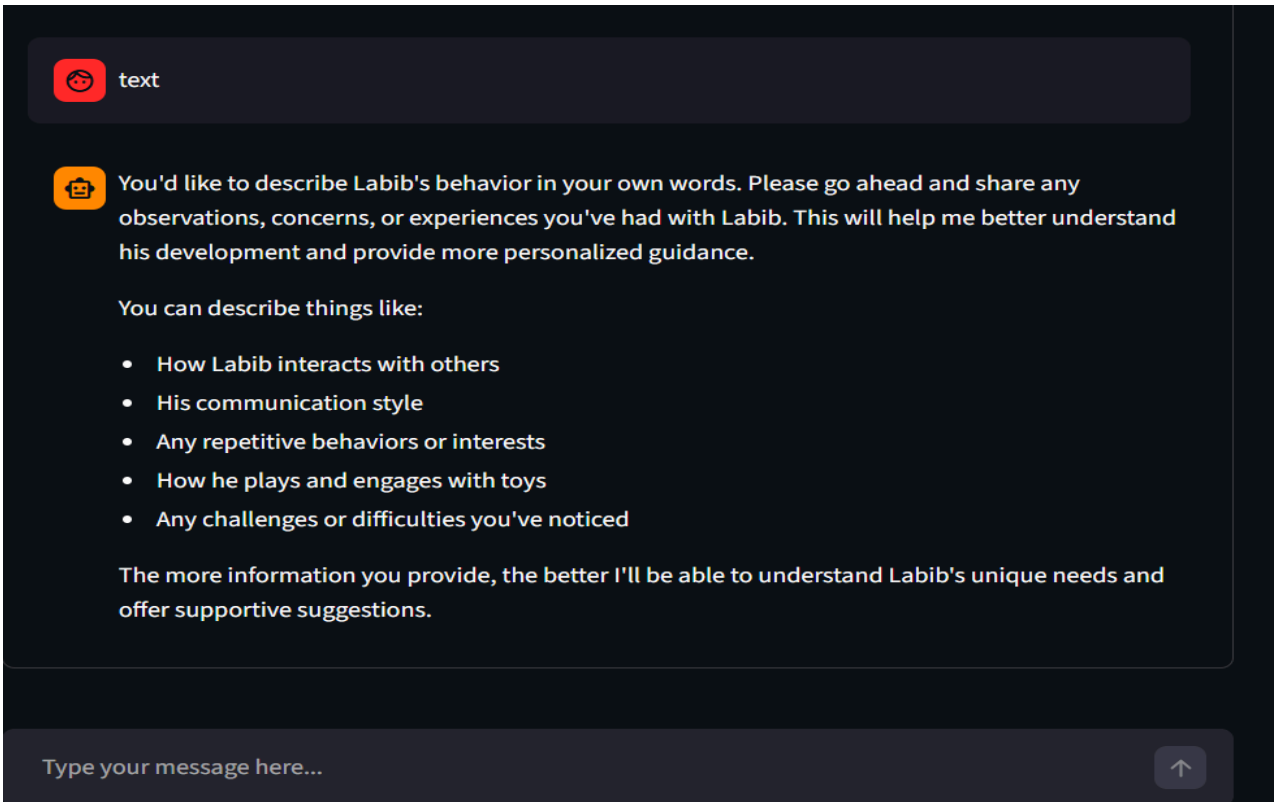
Following data entry, the system generates a result.



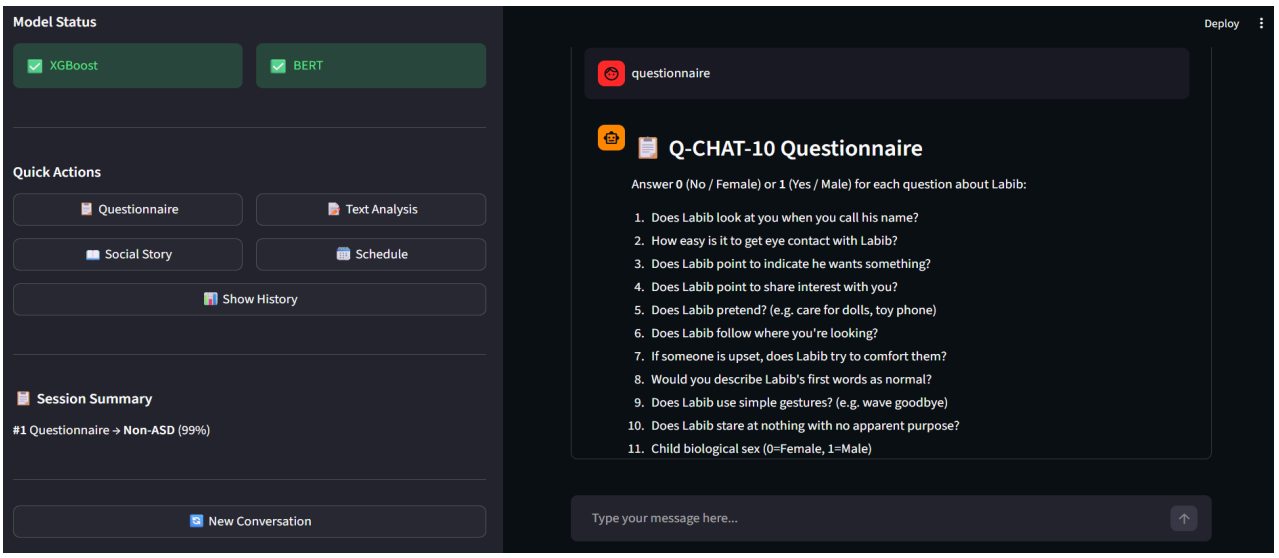
Users may then request an interpretation of the results or guidance on subsequent actions.



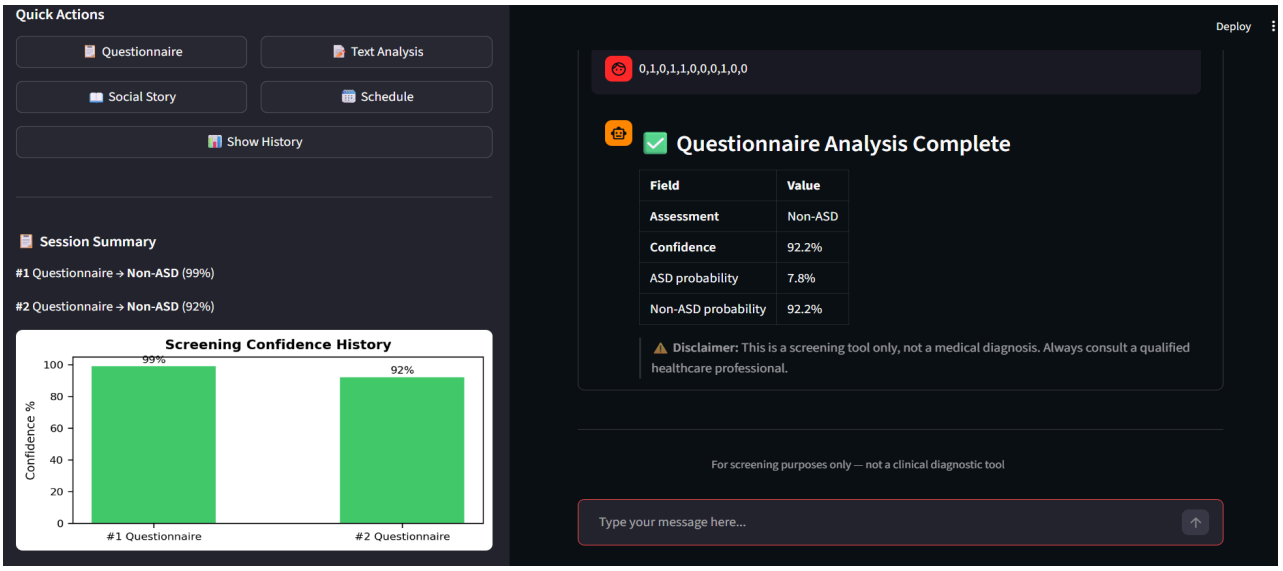
All assessments are logged within the session history for future reference



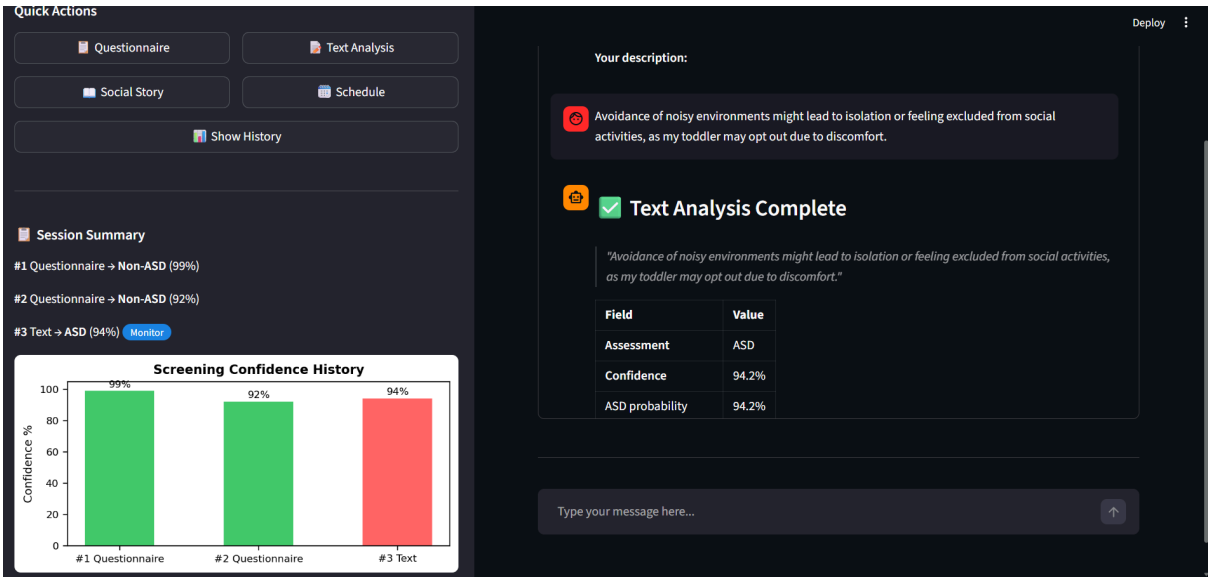
Alternatively, the user can select the text-based method.



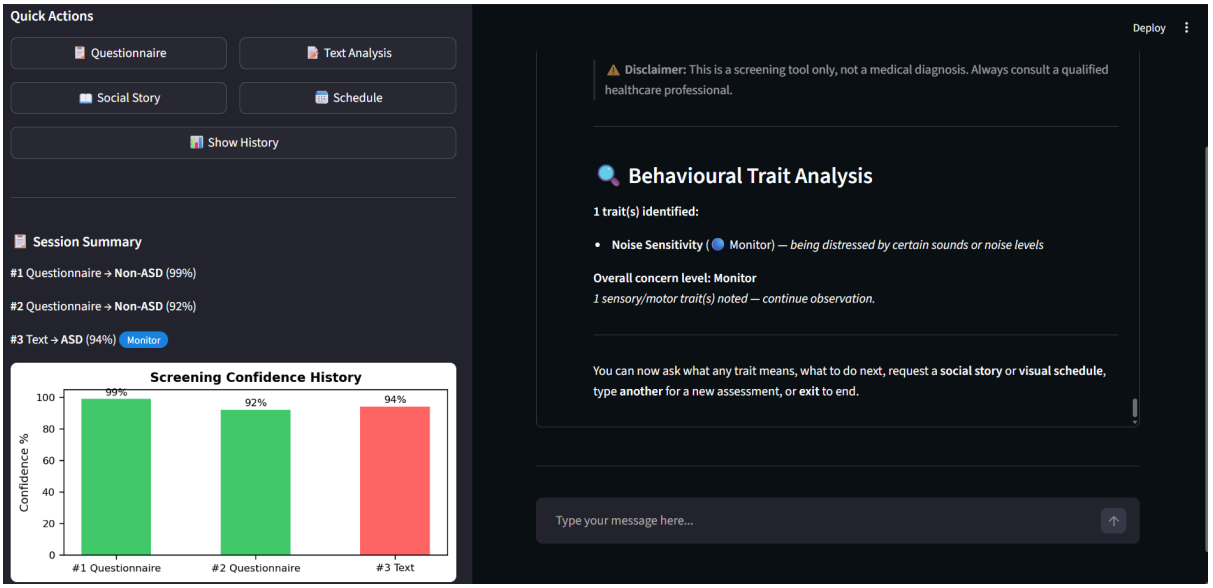
The user also has the option to repeat the questionnaire method.



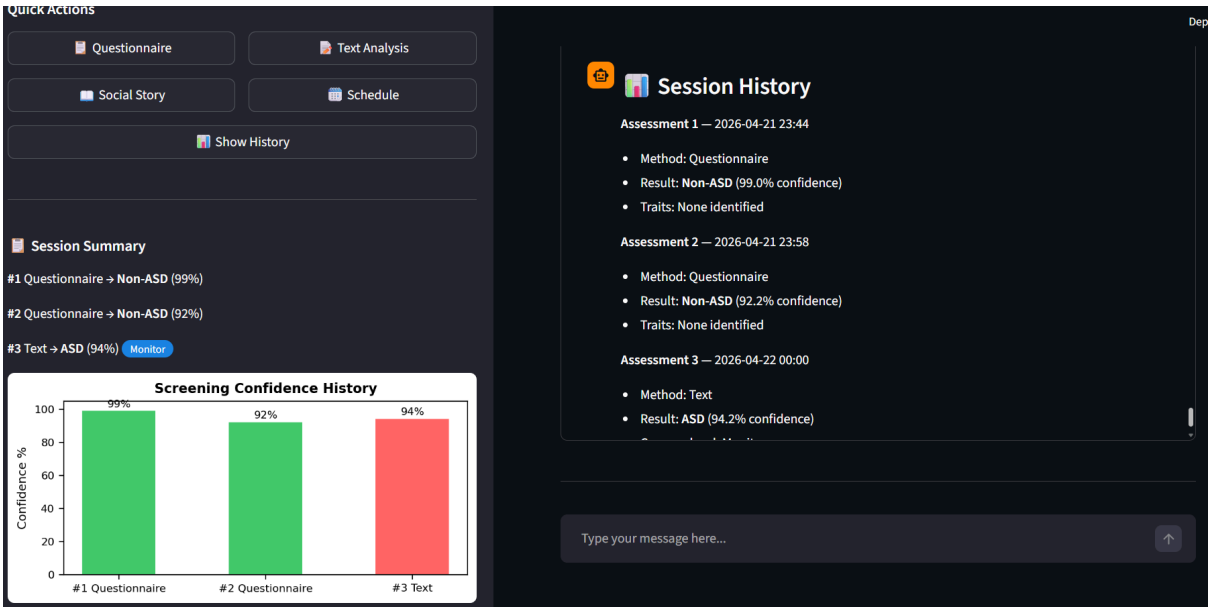
If the questionnaire is taken again, the system provides a new result and updates the Screening Confidence History bar chart in the sidebar.



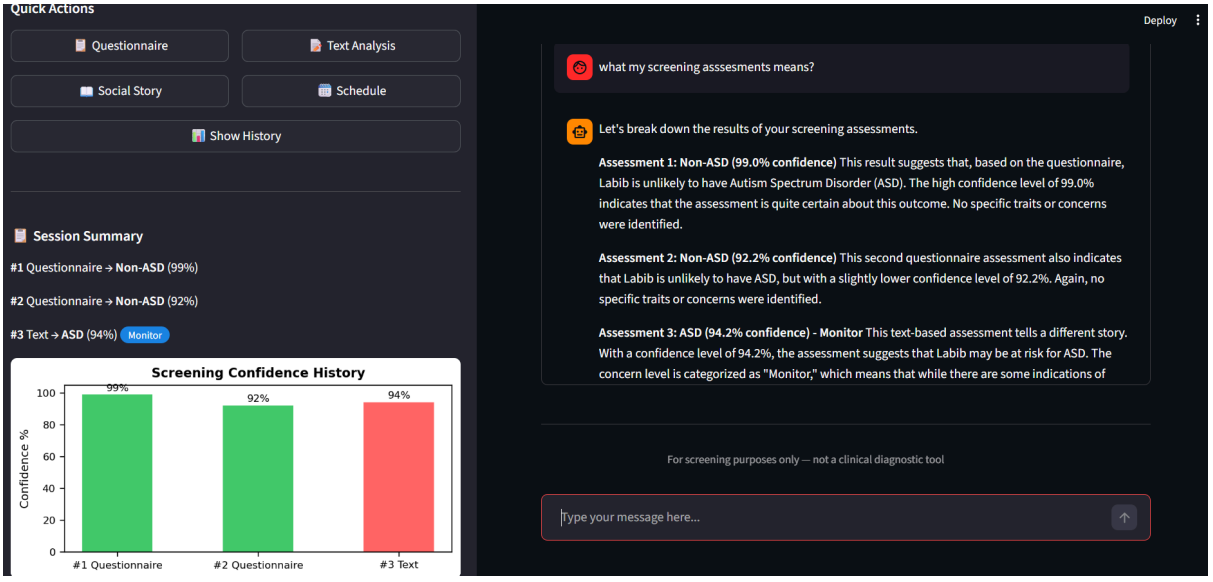
When the text method is selected, its results are similarly reflected and updated in the bar chart.



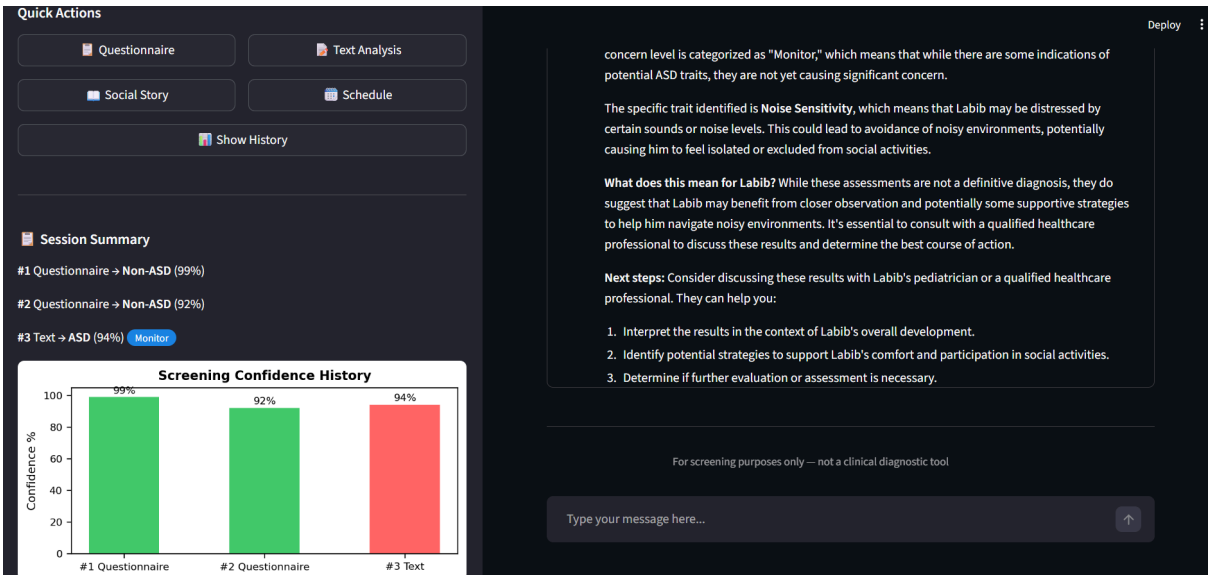
Additionally, the text method identifies specific behavioral traits within the provided description.



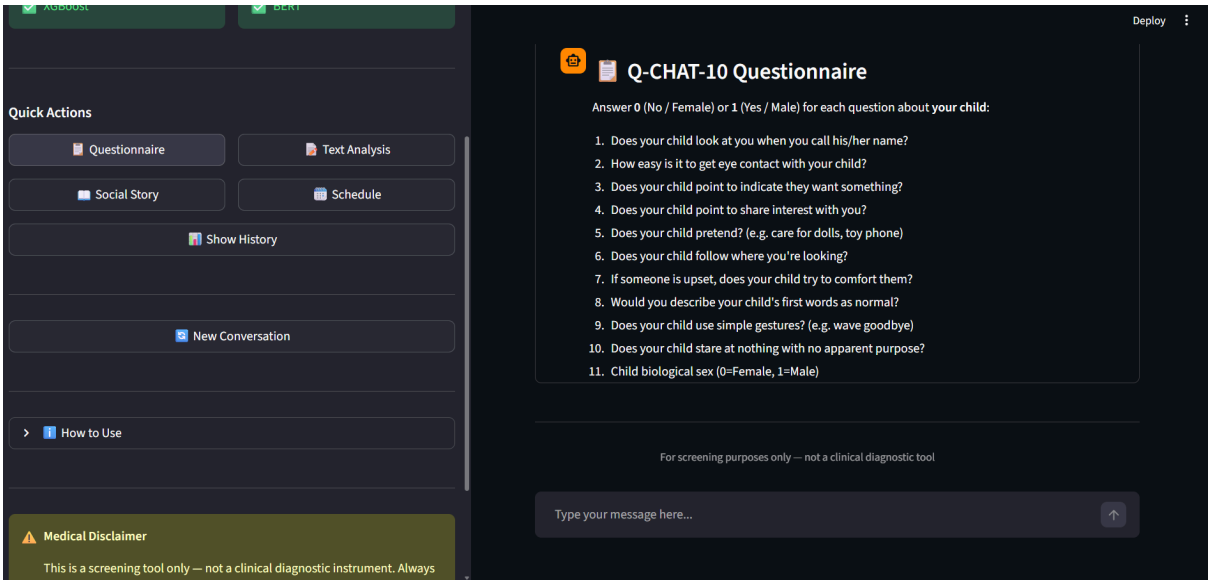
By clicking Session History in the sidebar, the user can review the results of all three assessments.



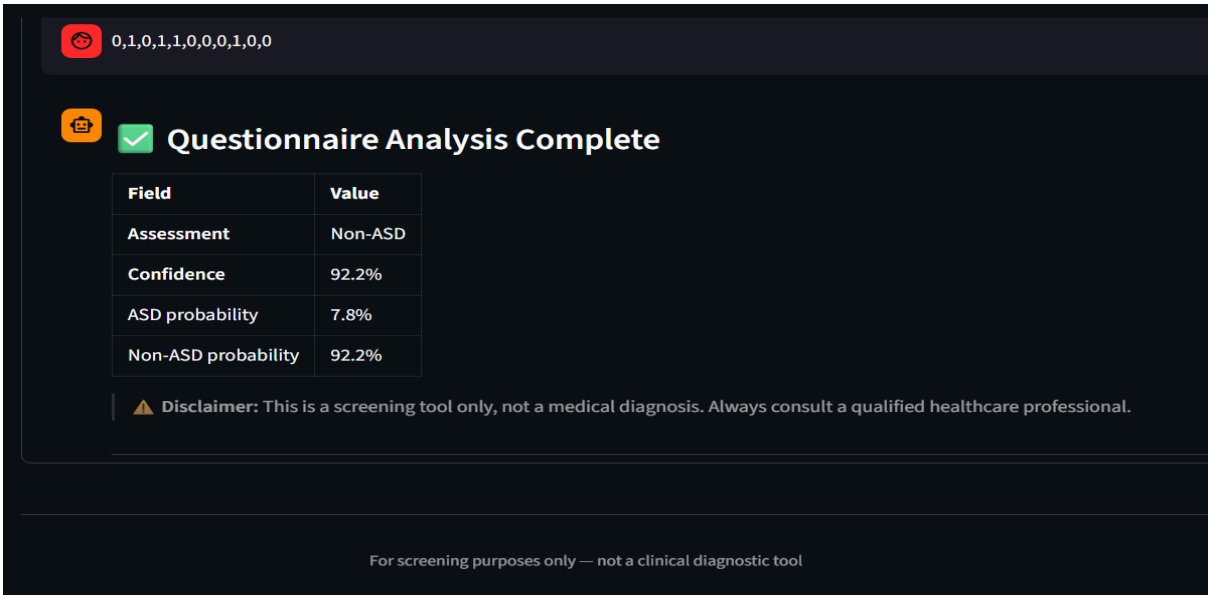
The user can then ask the AI to interpret what these combined results mean in relation to the child's previously provided description.



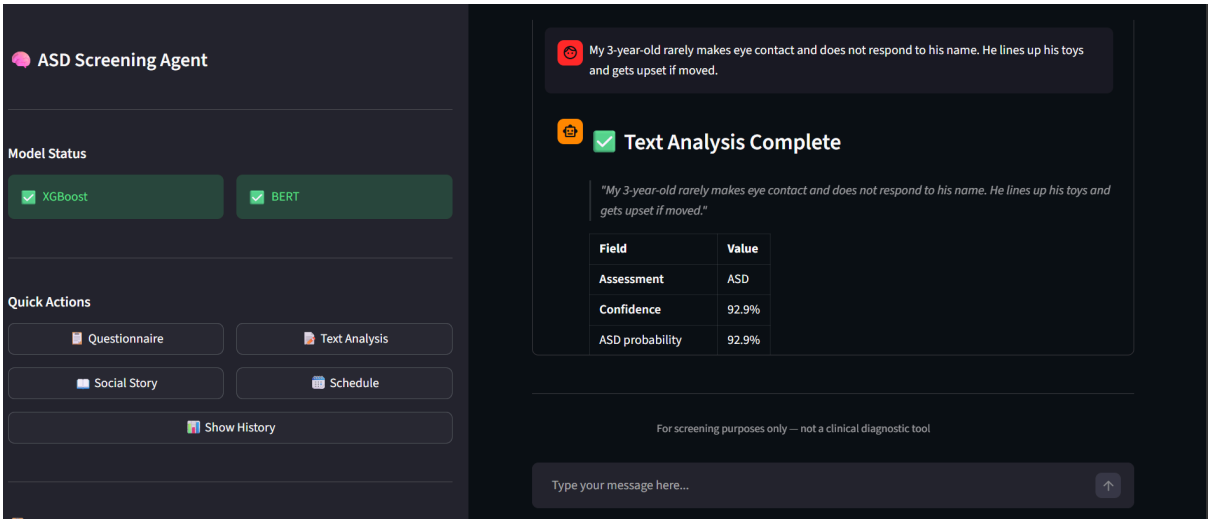
And it also suggests next steps

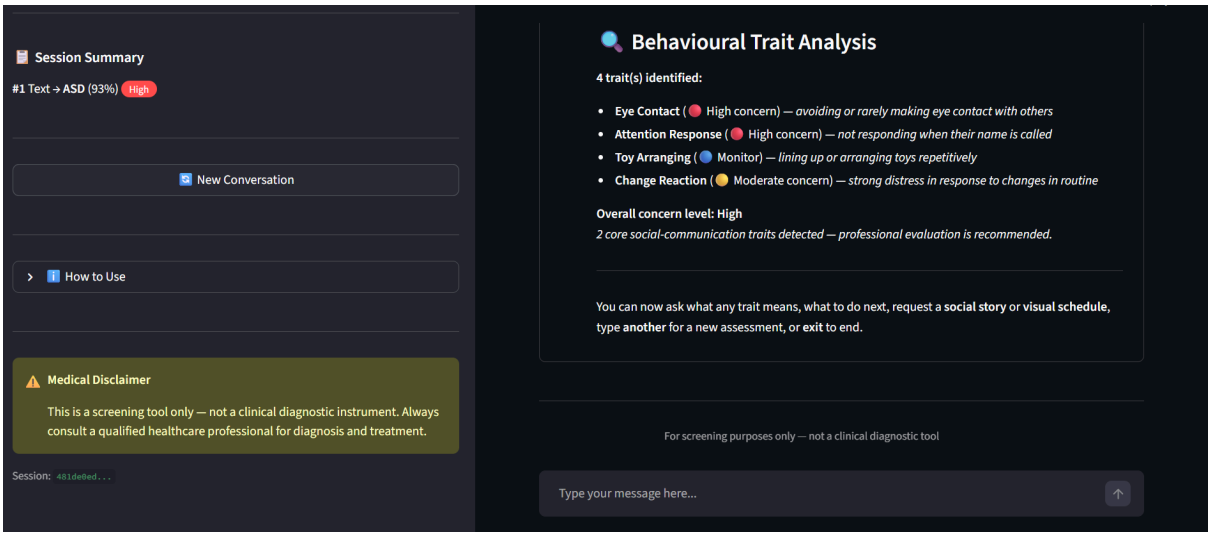


Users can also perform a Quick Analysis via the sidebar without providing personalized details.

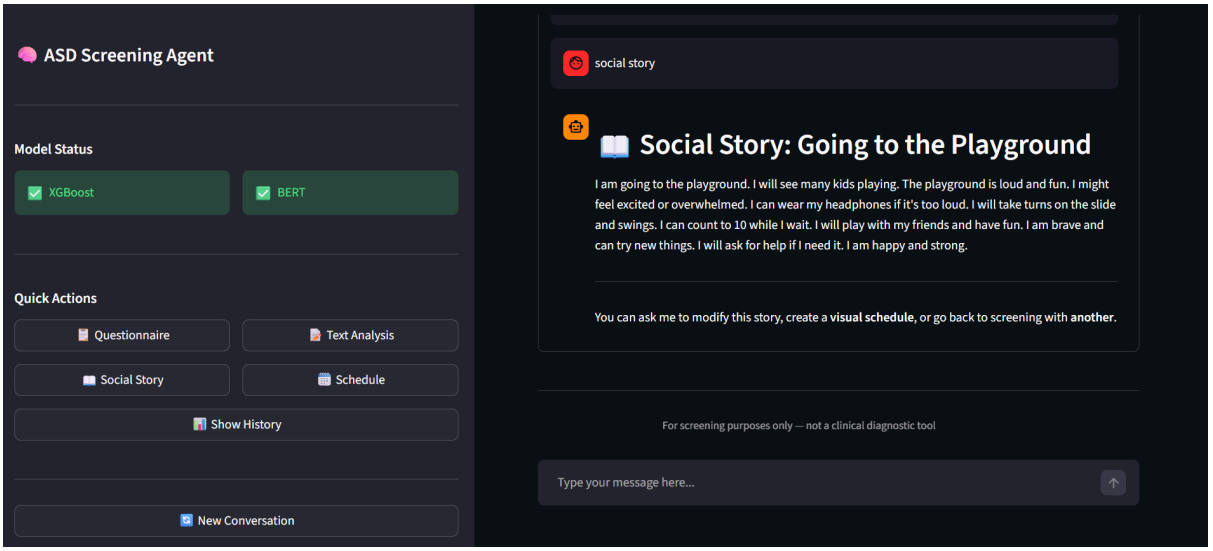


Results of selected method.





Results of selected method.



The AI agent can share Social Stories to support children by explaining concepts through storytelling.



Furthermore, the agent can generate schedules, such as a child's morning routine, to provide structured daily support.

Chapter 6: Conclusion

6.1 Overall Contributions

This project designed, implemented, and validated an AI Agent Framework for Autism Trait Detection in Children that makes the following substantive contributions to the state of the art.

The framework delivers dual-modality ASD screening within a single conversational session, integrating an XGBoost classifier achieving 98.6% accuracy and 0.9997 AUC-ROC on the Q-CHAT-10 structured questionnaire modality, and a fine-tuned BERT classifier achieving 88% accuracy and 0.96 AUC-ROC on the free-text behavioural description modality. Together, these two modalities serve a broader population of caregivers than any single-modality system.

The framework implements a verified architectural anti-hallucination boundary through LangGraph agent node separation, ensuring that no ASD classification statement is generated by the language model. All diagnostic outputs are produced exclusively by the trained ML classifiers. This represents a principled engineering response to the documented hallucination risk in medical AI applications.

A 14-trait behavioural taxonomy aligned with DSM-5 diagnostic domains is implemented and integrated into the free-text modality's output, providing interpretable trait-level explanation accompanied by a three-tier concern severity score calibrated against DSM-5 diagnostic weighting.

Semantically conservative data augmentation using protected-vocabulary synonym replacement and contextual BERT MLM augmentation was implemented, validated, and used to triple the BERT training corpus from 232 to 696 samples while preserving clinical vocabulary fidelity, demonstrating a practical approach to fine-tuning under data scarcity.

Companion capabilities personalised social story generation with FAISS semantic retrieval, visual schedule generation, child profile management, session history tracking, and sensory trigger detection transform the system from a one-shot screening tool into an ongoing caregiver companion grounded in evidence-based ASD intervention practice.

6.2 Limitations and Future Work

Limitation 1 : Small BERT training corpus. The fundamental ceiling imposed by 332 samples is reflected in the wide Wilson confidence interval [76.1%, 94.5%] on the 50-sample test set. Future work should pursue clinical partnership with paediatric developmental units to collect larger, professionally annotated text datasets, or explore data synthesis from clinical case notes using privacy-preserving techniques.

Limitation 2 : LLM-based trait extraction. The 14-trait identification relies on a constrained LLM call rather than a dedicated multi-label classifier, because the Mendeley dataset is too small to train one reliably per class. Future work should collect sufficient per-trait labelled examples to replace the LLM extraction with a supervised BERT multi-label classifier, eliminating LLM dependence from the trait output entirely.

Limitation 3 : English-language constraint. Both datasets are in English and the BERT tokeniser was trained on English text. The tool is currently inaccessible to the majority of the global ASD-affected population. Future work should investigate multilingual BERT (mBERT) or XLM-RoBERTa fine-tuning for Bengali, Arabic, Spanish, and other high-prevalence-ASD languages.

Limitation 4 : Absence of prospective clinical validation. The system has been evaluated on held-out splits of publicly available datasets, not on prospective real-world data from a clinical deployment. A formal clinical validation study in partnership with a paediatric developmental service is required before any clinical deployment recommendation can be made.

Limitation 5 : LLM API dependency. The orchestrating language model (gpt-oss-120b) requires API access with per-token costs and latency that may be prohibitive in low-resource settings. Future work should investigate locally-deployable smaller LLMs (Mistral-7B, Llama-3-8B) as API-free alternatives.

Limitation 6 : No formal usability evaluation. The system has not been evaluated by actual caregivers of children with ASD in a structured usability study. Future work should conduct formal usability testing including WCAG accessibility compliance evaluation, caregiver satisfaction assessment, and field testing in community health settings.

6.3 Business Model Analysis

The system occupies the intersection of digital health, AI assistive technology, and early childhood intervention, a market projected to grow at a compound annual growth rate exceeding 7% through 2030.

Value Proposition. For families in low-resource or remote settings, the system provides free-at-point-of-use ASD screening that would otherwise require a specialist appointment costing hundreds of dollars and weeks of waiting time. For primary care paediatricians, it functions as a validated digital pre-screening tool that can be completed asynchronously before the clinical visit. For early childhood educators and speech-language therapists, it provides automated generation of personalised social stories and visual schedules that currently require significant manual time to produce.

Revenue Model. A freemium architecture is most appropriate. The core dual-modality screening remains permanently free to ensure accessibility. A premium subscription tier (approximately $5–10/month for families, $50–100/month for clinics) unlocks unlimited companion content generation, multi-child profile management, long-term progress tracking with visualisation, and integration with family calendar and reminder systems. A B2B institutional licensing model targeting school districts, NGOs operating in low-resource settings, and developmental paediatric networks provides the primary sustainable revenue stream.

Competitive Differentiation. No currently deployed commercial tool combines dual-modality validated ASD screening, LLM-driven companion features, FAISS-based material retrieval, architectural anti-hallucination protection, and personalised child profile management in a single integrated conversational system. The open-source LangGraph architecture and use of publicly available pre-trained models keep marginal serving costs low.

Ethical and Regulatory Considerations. The system must be positioned as a screening tool, not a clinical diagnostic instrument in all user communications, terms of service, and regulatory filings. Any expansion toward clinical deployment in markets such as Bangladesh, the United Kingdom, or the United States would require engagement with the applicable regulatory framework (Directorate General of Drug Administration, Medicines and Healthcare products Regulatory Agency, and FDA Software as a Medical Device guidance, respectively). Data privacy compliance with local personal data protection legislation is mandatory from day one of deployment.

6.4 Final Remarks

The proposed framework demonstrates the potential of combining machine learning, natural language processing, and agent-based system design to address real-world healthcare challenges. By prioritising accessibility, reliability, and usability, this work contributes to the development of scalable tools for early ASD screening and support.

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Appendix

Program Outcomes

PO1 (Engineering Knowledge):
This project applies mathematical, scientific, and engineering principles to develop a deep learning–based lung histopathology classification system. Multiple CNN architectures (ResNet50, DenseNet121, EfficientNetB0, MobileNetV2, MobileNetV3Small) and ensemble techniques such as weighted voting, stacking, and boosting demonstrate strong grounding in machine learning, computer vision, and engineering fundamentals.

PO2 (Problem Analysis):
The project identifies key challenges in histopathology images, such as stain variability, magnification differences, class imbalance, and patient-level generalization. The introduction of patient-wise splitting and ensemble fusion techniques reflects systematic problem analysis and solution formulation.

PO3 (Design/Development of Solutions):
A dynamic hybrid ensemble system is designed, allowing flexible combinations of CNN models without retraining. Weighted voting, stacking, and boosting pipelines provide alternative diagnostic mechanisms and optimize prediction reliability.

PO4 (Investigation):
The project conducts extensive experimentation across five CNN backbones, three boosting models, and multiple ensemble strategies. It evaluates image-level and patient-level performance, validating findings with controlled experiments and comparative analysis.

PO5 (Modern Tool Usage):
Modern tools and frameworks such as TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Matplotlib, XGBoost, LightGBM, CatBoost, and Google Colab GPUs are used for model development, training, and evaluation. FastAPI is used for deployment testing.

PO6 (Engineer and Society):
The system contributes to healthcare by assisting in early lung cancer detection and supporting clinical decision-making. The dataset contains no personal patient information, ensuring ethical compliance and protecting privacy.

PO7 (Environment and Sustainability):
Lightweight CNN backbones and efficient ensemble strategies are explored to reduce computational cost, making the system sustainable for real-world medical environments with limited hardware resources.

PO8 (Ethics):
Ethical standards regarding dataset usage, privacy preservation, and transparent reporting are maintained. All dataset sources are publicly available and anonymized.

PO9 (Individual and Teamwork):
The project combines teamwork in experimentation, documentation, coding, and analysis. Group members collaborated in structured workflows, version control, and peer review.

PO10 (Communication):
The research communicates results clearly through graphs, tables, and structured descriptions. The FastAPI-based demo enables intuitive communication of system behavior to technical and non-technical audiences.

PO11 (Project Management and Finance):
The project was executed within a planned timeline using free or academic tools (Google Colab, Python libraries). Efficient use of cloud resources and proper task distribution ensured cost-effective completion.

PO12 (Life Long Learning):
The project reflects continuous learning in deep learning, ensemble modeling, deployment frameworks, and medical imaging. It encourages future extension into clinical applications and research advancements.

Mapping of Course and Program Outcomes

Addressing of COs, Knowledge Profile (K), and Complex Engineering Problem(EP):

Program Outcome (PO)	**PO Description**	**Project Evidence**
PO1	Engineering Knowledge: Apply knowledge of mathematics, science, and engineering fundamentals	Applied probability theory (AUC-ROC, Wilson confidence intervals), information theory (point-biserial correlation, chi-square hypothesis testing), gradient boosting mathematics, and transformer attention mechanism principles throughout all model development activities.
PO2	Problem Analysis: Identify and analyse complex engineering problems	Conducted systematic literature review identifying the dual-modality gap and anti-hallucination design requirement; formulated the problem as a multi-objective engineering challenge requiring simultaneous satisfaction of accuracy, safety, accessibility, and deployability constraints.
PO3	Design/Development of Solutions: Design systems meeting specified needs	Designed the complete LangGraph multi-agent architecture, FSM routing, dual-modality pipeline, trait taxonomy, and companion capability layer, with explicit design decisions justified against safety (anti-hallucination), accessibility (parent-answerable features), and ethical (ethnicity exclusion) criteria.

PO4	Investigation: Conduct experiments and interpret data	Designed and executed comparative ML experiments (four-model comparison, 5-fold CV, AUC-ROC analysis, Wilson CI computation); conducted chi-square and point-biserial correlation analyses for feature selection; performed two-pass augmentation strategy validation.
PO5	Modern Tool Usage: Use modern engineering tools effectively	Applied XGBoost 2.0, HuggingFace Transformers BERT, LangGraph, FAISS, sentence-transformers, nlpaug, PyTorch, Streamlit, and Google Colab GPU infrastructure; justified each tool selection on technical and practical grounds; acknowledged limitations of each.
PO6	The Engineer and Society: Assess societal and health impacts	Implemented the anti-hallucination boundary as a safety engineering decision; excluded ethnicity as a feature on ethical grounds; included a medical disclaimer in all user-facing output; discussed regulatory frameworks (FDA, MHRA, DGDA) in the business model analysis.
PO7	Environment and Sustainability: Understand sustainable impact	Analysed the scalability and zero-marginal-cost-per-screening nature of the system as a mechanism for addressing the structural inequality in ASD screening access between high-income and low-resource settings.
PO8	Ethics: Apply ethical principles in engineering practice	Rejected ethnicity as a feature on ethical grounds despite its statistical significance; implemented the medical disclaimer by design; committed to transparency regarding the screening-vs-diagnosis boundary in all system outputs and user documentation.

PO9	Individual and Team Work: Work effectively in teams and multidisciplinary settings	Executed the capstone project as an individual integrating knowledge across machine learning, natural language processing, software engineering, human-computer interaction, and clinical domain knowledge simultaneously.
PO10	Communication: Communicate effectively with diverse audiences	Produced this technical thesis report; delivered a poster presentation at the capstone symposium; designed the Streamlit interface for non-specialist caregiver communication; documented all code with inline explanations of clinical and methodological decisions.
PO11	Project Management and Finance: Apply management principles	Section 6.3 provides a complete business model analysis covering value proposition, revenue model options (freemium, B2B), competitive differentiation, cost structure, and regulatory compliance planning across three jurisdictions.
PO12	Life-long Learning: Engage in continuous learning	Section 6.2 frames six specific future research directions; engagement with current literature on LLMs in medicine [2], multicultural ASD ML [13], text augmentation [4], and RAG systems [22] reflects active monitoring of the relevant research frontier.